

Structural basis of EHEP-mediated offense against phlorotannin-induced defense from brown algae to protect *akuBGL* activity

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Xiaomei Sun, Yuxin Ye, Naofumi Sakurai, Hang Wang, Koji Kato, Jian Yu, Keizo Yuasa, Akihiko Tsuji, Min Yao 

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan • Graduate School of Bioscience and Bioindustry, Tokushima University, Tokushima, Japan

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Abstract

The defensive-offensive associations between algae and herbivores determine marine ecology. Brown algae utilize phlorotannin as their chemical defense against the predator *Aplysia kurodai*, which uses β -glucosidase (*akuBGL*) to digest the laminarin in algae to glucose. Moreover, *A. kurodai* employs *Eisenia* hydrolysis-enhancing protein (EHEP) as an offense to protect *akuBGL* activity from phlorotannin inhibition by precipitating phlorotannin. To underpin the molecular mechanism of this digestive-defensive-offensive system, we determined the structures of apo and tannic-acid (TNA, a phlorotannin-analog) bound form of EHEP, as well as apo *akuBGL*. EHEP consisted of three peritrophin-A domains formed in a triangle and bound TNA in the center without significant conformational changes. Structural comparison between EHEP and EHEP-TNA led us to find that EHEP can be resolubilized from phlorotannin-precipitation at an alkaline pH, which reflects a requirement in the digestive tract. *akuBGL* contained two GH1 domains, only one of which conserved the active site. Combining docking analysis, we propose the mechanisms by which phlorotannin inhibits *akuBGL* by occupying the substrate-binding pocket, and EHEP protects *akuBGL* against the inhibition by binding with phlorotannin to free the *akuBGL* pocket.

eLife assessment

This **important** study presents **convincing** evidence on how the sea slug *Aplysia kurodai* optimizes its digestion of brown algae, in a classical predator-prey 'arms race' at the molecular level. The experimental protein structures and enzyme assays provide support for the claims of how *A. kurodai* avoids inhibition by algal compounds, and also hold promise for biotechnological applications.

Over millions of years of evolution, predators have successfully coevolved with their prey to maintain an ecological balance ¹. In marine habitats, interactions between algae and marine herbivores dominate marine ecosystems. Most algae are consumed by marine herbivores ². They produce secondary metabolites as a chemical defense to protect themselves against predators. In brown algae *Eisenia bicyclis*, laminarin is a major storage carbohydrate, constituting 20%–30% of algae dry weight. The sea hare *Aplysia kurodai*, a marine gastropod, preferentially feeds on the *E. bicyclis* with its 110 and 210 kDa β -glucosidases (*akuBGLs*), hydrolyzing the laminarin and releasing large amounts of glucose. Interestingly, such a feeding strategy has attracted attention for producing glucose as a renewable biofuel source ³. However, to protect themselves against predators, brown algae produce phlorotannin, a secondary metabolite, thereby reducing the digestion by *A. kurodai* by inhibiting the hydrolytic activity of *akuBGLs*. As the 110 kDa *akuBGL* is more sensitive to phlorotannin than the 210 kDa BGL ⁴, we focused on the 110 kDa *akuBGL* in this study (hereafter, *akuBGL* refers to 110 kDa *akuBGL*).

To counteract the antipredator adaptations of algae, herbivores use diverse approaches, such as detoxification, neutralization, defense suppression, and physiological adaptations ⁵. *A. kurodai* inhibits the phlorotannin-defense of brown algae through *Eisenia* hydrolysis-enhancing protein (EHEP), a protein from their digestive system that protects *akuBGL* activity from phlorotannin inhibition ⁴. Previous studies have shown that incubating *E. bicyclis* with *akuBGL* in the presence of EHEP results in increased glucose production because EHEP binds to phlorotannin and forms an insoluble complex ⁴.

The *akuBGL*–phlorotannin/laminarin–EHEP system exemplifies the digestion process of *A. kurodai* as well as the defense and antidefense strategies between *E. bicyclis* and *A. kurodai*. Although the defense/antidefense strategy has been established, the detailed molecular mechanism of this interplay remains unknown. Further, phlorotannin inhibition hinders the potential application of brown algae as feedstocks for enzymatically producing biofuel from laminarin. Thus, understanding the underlying molecular mechanisms will be beneficial for the application of this system in the biofuel industry.

Despite the potential use of laminarin hydrolytic enzymes in the biofuel industry, only a few BGLs of glycoside hydrolases belonging to the GH3 and GH1 family are known to hydrolyze laminarin (e.g., *Talaromyces amestolkiae* BGL ⁶, *Ustilago esculenta* BGL ⁷, and *Vibrio campbellii* BGL ⁸ from the GH3 family and *Saccharophagus degradans* 2-40^T BGL ⁹ from the GH1 family). GH3 is a multidomain enzyme family characterized by N-terminal (β/α)₈ (NTD) and C-terminal (β/α)₆ (CTD) domains, with or without auxiliary domains ¹⁰; the nucleophile aspartate and the acid/base glutamate residues exist in the NTD and CTD, respectively. In contrast, the members of the GH1 family generally share a single (β/α)₈-fold domain (hereafter referred to as GH1 domain [GH1D]), and the two glutamic acid catalytic residues are located in the carboxyl termini of β -strands 4 and 7. Therefore, the two families may use different substrate-recognition and catalytic mechanisms for laminarin. Intriguingly, although *akuBGL* possesses laminarin hydrolytic activity and belongs to the GH1 family, its molecular weight is considerably greater than that of other GH1 members. Sequence analysis has indicated that *akuBGL* consists of ≥ 2 GH1Ds. Because no structural information of BGL active on polysaccharides is available, the catalytic mechanism toward laminarin is unclear.

There is limited information on EHEP, a novel cysteine-rich protein (8.2% of the amino acid content), because no structural or functional homologous protein exists in other organisms. EHEP was predicted to consist of three peritrophin-A domains (PADs) with a cysteine-spacing pattern of CX₁₅CX₅CX₉CX₁₂CX₅₋₉C. The PADs consist of peritrophic matrix proteins, which have been proposed to play an important role in detoxifying ingested xenobiotics ¹¹. For instance, *Aedes aegypti* intestinal mucin 1 (*AeIMUC1*) consists of a signal peptide followed by three PADs with an

intervening mucin-like domain; its expression is induced by blood feeding. *AeIMUC1*-mediated blood detoxification during digestion is completed by binding to toxic heme molecules.¹² Despite the similar domain organization of EHEP and *AeIMUC1*, their function and binding partner are completely different, implying their different characteristics. However, the characteristics of the EHEP-phlorotannin insoluble complex remain unknown; moreover, it remains unclear why and how EHEP protects *akuBGL* from phlorotannin inhibition.

In this study, we determined the structures of apo and tannic acid (TNA, phlorotannin-analog) bound-form of EHEP (EHEP, EHEP-TNA), as well as *akuBGL*; all isolated from *A. kurodai*. The structure of EHEP consists of three PADs arranged in a triangle shape, with TNA bound at the surface of the triangle center. A structural comparison of EHEP and EHEP-TNA revealed no significant changes in conformation upon TNA binding, implying that EHEP maintains its structure when precipitated with TNA. Then, we found the conditions to resolubilize EHEP-TNA precipitate for EHEP recycling. The obtained *akuBGL* structure suggests that only one GH1D (GH1D2) possesses laminarin hydrolytic activity; subsequently, ligand-docking experiments demonstrated that TNA/phlorotannin has a higher docking score than laminarin. Our results revealed the mechanisms by which EHEP protects *akuBGL* from phlorotannin inhibition and phlorotannin inhibits the hydrolytic activity of *akuBGL*, providing structural support for the potential application of brown algae for biofuel production.

Results

Effects of TNA on *akuBGL* activity with or without EHEP

Phlorotannin, a type of tannin, is a chemical defense metabolite of brown algae. It is difficult to isolate a compound from phlorotannins because they are a group of polyphenolic compounds with different sizes and varying numbers of phloroglucinol units¹³, such as eckol, dieckol, and so on¹⁴. Previous studies have reported that the phlorotannin-analog TNA has a comparable inhibition effect on *akuBGL* to that of phlorotannin⁴. Hence, we used TNA instead of a phlorotannin to explore phlorotannin binding with EHEP and *akuBGL*. This activity assay system involves multiple equilibration processes: *akuBGL* ⇌ substrate, *akuBGL* ⇌ TNA, and EHEP ⇌ TNA. First, we confirmed TNA inhibition of *akuBGL* activity and clarified the protective effects of EHEP from TNA inhibition. The inhibition experiments showed that the galactoside hydrolytic activity of *akuBGL* decreased with increasing TNA concentration, indicating that TNA inhibits *akuBGL* activity in a dose-dependent manner (**Fig. 1A**, **Fig. S1A**). Approximately 70% *akuBGL* activity was inhibited at a TNA concentration of 40 μM. Moreover, protection ability analysis revealed that EHEP protects *akuBGL* activity from TNA inhibition in a dose-dependent manner, as indicated by the recovery of the inhibited *akuBGL* activity with increasing EHEP concentration (**Fig. 1B**, **Fig. S1B**). Further, approximately 80% of *akuBGL* activity was recovered at an EHEP concentration of 3.36 μM.

Overall structure of EHEP

Considering the lack of known homologous proteins of EHEP, we determined the structure of EHEP using the native-SAD method at a resolution of 1.15 Å, with a R_{work} and R_{free} of 0.18 and 0.19, respectively (**Table 1**). The residues A21–V227 in purified EHEP (1–20 aa were cleaved during maturation) were built, whereas two C-terminal residues were disordered. The structure of EHEP consists of three PADs: PAD1 (N24–C79), PAD2 (I92–C146), and PAD3 (F164–C221), which are linked by two long loops, LL1 (Q80–N91) and LL2 (R147–G163), and arranged in a triangle shape (**Fig. 2A**). These three PADs share a similar structure, with a root-mean-squared difference (RMSD) of 1.065 Å over 46 Ca atoms and only ~20.3% sequence identity (**Figs. 2B** and **2C**). The

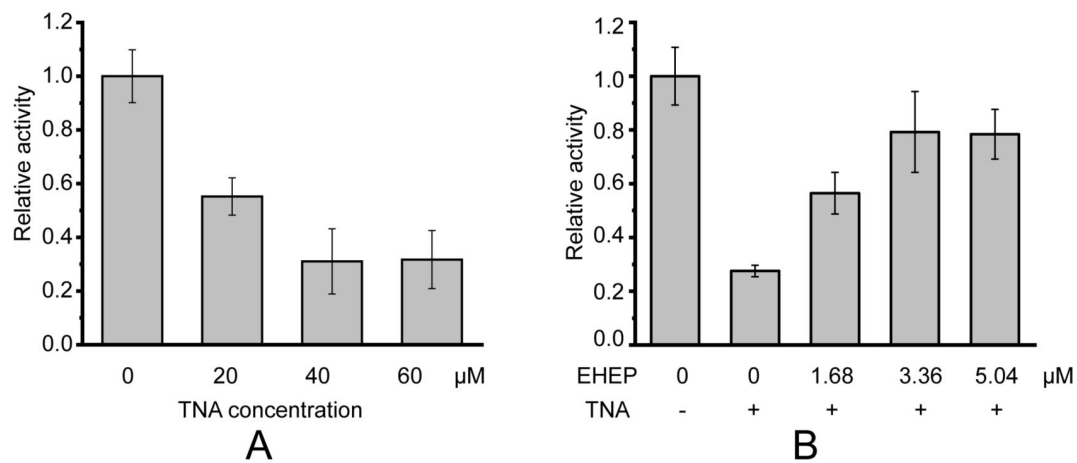


Figure 1.

Galactoside hydrolytic activity of *akuBGL* toward ortho-Nitrophenyl- β -galactoside. Activity (%) is shown as the fold increase relative to *akuBGL* without the addition of TNA nor EHEP. (A) The hydrolytic activity of *akuBGL* (0.049 μ M) with TNA at different concentrations. (B) The hydrolytic activity of *akuBGL* (0.049 μ M) with 40 μ M TNA and EHEP at different concentrations. The average and standard deviation of the relative activity were estimated from three independent replicates (N = 3).

three PADs share a canonical CBM14 fold consisting of two β -sheets containing three N-terminal and two C-terminal antiparallel β -strands. Additionally, two small α -helices were appended to the N- and C-terminus in PAD1 and PAD3 but not in PAD2 (**Fig. 2B** [↗](#)).

Although the Dali server [15](#) [↗](#) did not provide similar structures using the overall structure of EHEP as the search model, six structures showed similarities with a single PAD of EHEP. These structures were the members of the PAD family, including the chitin-binding domain of chitotriosidase (PDB ID 5HBF) [16](#) [↗](#), avirulence protein 4 from *Pseudocercospora fuligena* [PfAvr4 (PDB ID 4Z4A)] [17](#) [↗](#) and *Cladosporium fulvum* [CfAvr4 (PDBID 6BN0)] [18](#) [↗](#), allergen Der p 23 (PDB ID 4ZCE) [19](#) [↗](#), tachytitin (PDB ID 1DQC) [20](#) [↗](#), and allergen Blot 12 (PDB ID 2MFK), with Z-scores of 4.7–8.4, RMSD values of 1.2–2.8 Å, and sequence identity of 19%–37%. The highest sequence disparity was detected in PAD2, whereas the greatest structural differences were noted in PAD3. The C^{No1}X₁₅C^{No2}X₅C^{No3}X₉C^{No4}X₁₂C^{No5}X₅₋₉C^{No6} [↗](#) motif (superscripts and subscripts indicate the cysteine number and number of residues between adjacent cysteines, respectively) in each PAD of EHEP formed three disulfide bonds between the following pairs: C^{No1}–C^{No3}, C^{No2}–C^{No6}, and C^{No4}–C^{No5} (**Fig. 2B** [↗](#)). Such rich disulfide bonds may play a folding role in the structural formation of EHEP, with >70% of the backbone in a loop conformation. A similar motif with disulfide bonds was observed in tachytitin [20](#) [↗](#), PfAvr4 [17](#) [↗](#), CfAvr4 [18](#) [↗](#), and the chitin-binding domain of chitinase [16](#) [↗](#). Although these proteins share a highly conserved core structure, they have different biochemical characteristics. For example, the chitin-binding domain of human chitotriosidase, Avr4 and tachytitin possess chitin-binding activity, but the critical residues for chitin-binding are not conserved [16](#) [↗](#), [18](#) [↗](#), [21](#) [↗](#), indicating that they employ different binding mechanisms. In contrast, EHEP and allergen Der p 23 do not possess chitin-binding activity [4](#) [↗](#), [19](#) [↗](#). Thus, the PAD family may participate in several biochemical functions.

Modification of EHEP

Consistent with the molecular weight results obtained using MALDI-TOF MS [22](#) [↗](#), the apo structure2 (1.4 Å resolution) clearly showed that the cleaved N-terminus of Ala21 underwent acetylation (**Fig. S2A** [↗](#)), demonstrating that EHEP is acetylated in *A. kurodai* digestive fluid. N-terminal acetylation is a common modification in eukaryotic proteins. Such acetylation is associated with various biological functions, such as protein half-life regulation, protein secretion, protein–protein interaction, protein–lipid interaction [23](#) [↗](#), metabolism, and apoptosis [24](#) [↗](#). Further, N-terminal acetylation may stabilize proteins [25](#) [↗](#). To explore whether acetylation affects the protective effects of EHEP on *akuBGL*, we used the *E.coli* expression system to obtain the unmodified recomEHEP (A21–K229). We measured the TNA-precipitating assay of recomEHEP. The results revealed that recomEHEP precipitated after incubation with TNA, at a comparable level to that of natural EHEP (**Fig. S2B** [↗](#)), indicating that acetylation is not indispensable for the phlorotannin binding activity and stabilization of EHEP. Future studies are warranted to verify the exact role of N-terminal acetylation of EHEP in *A.kurodai*.

TNA binding to EHEP

To understand the mechanism by which TNA binds to EHEP, we determined the structure of EHEP complexed with TNA (EHEP–TNA) using the soaking method. In the obtained structure, both 2F_o–F_c and F_o–F_c maps showed the electron density of 1,2,3,4,6-pentagalloylglucose, a core part of TNA missing the five external gallic acids (**Fig. 3A** [↗](#), **Fig. S3A** [↗](#)). Previous studies have shown that acid catalytic hydrolysis of TNA requires a high temperature of 130°C [26](#) [↗](#); even with a polystyrene-hollow sphere catalyst, a temperature of 80°C is required [27](#) [↗](#). Therefore, the five gallic acids could not be visualized in the EHEP–TNA structure most likely due to the structural flexibility of TNA.

The apo EHEP and EHEP–TNA structures were extremely similar, with an RMSD value of 0.283 Å for 207 Ca atoms (**Fig. S3B** [↗](#)). However, the superposition of the two structures showed a decrease of the loop part in EHEP–TNA. TNA binding caused a slight increase in the α -helix and β -

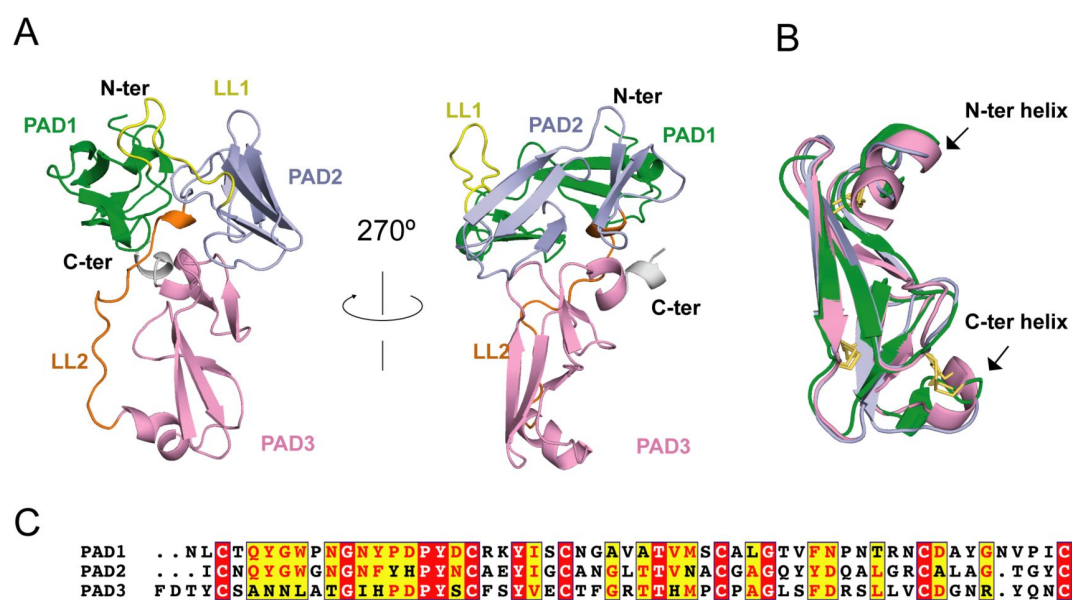


Figure 2.

EHEP structure. (A) Cartoon representation of EHEP. The three PAD domains are colored green, light blue, and pink, respectively. Linker long loop1 (LL1) and loop2 (LL2) are colored yellow and blue. (B) Structural superposition of the three PAD domains of EHEP. The three domains are colored as in (A). The disulfide bonds are shown as yellow sticks. (C) Sequence alignment of three PAD domains. Alignment was performed by CLUSTALW and displayed with ESPrpt3.

	EHEP1	EHEP2	EHEP-TNA	akuBGL
Data Collection				
Beamline	PF BL17A	Spring 8 BL-41XU	PF BL17A	PF BL1A
Wavelength (Å)	0.9800	1.0000	0.9800	1.0000
Resolution range (Å)	46.56–1.15 (1.20–1.15)	47.02–1.4 (1.45–1.4)	46.84–1.9 (1.97–1.9)	49.65–2.7 (2.80–2.7)
Space group	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$	$P6_2$
Unit-cell parameters a, b, c (Å)	42.2, 65.3, 66.5	40.6, 65.6, 67.5	42.5, 65.4, 67.2	91.7, 191.7, 12.6
Completeness (%)	93.4 (75.8)	99.30 (97.98)	99.9 (99.7)	99.9 (99.1)
Redundancy	6.6 (6.1)	6.4 (5.8)	6.4 (6.4)	10.7 (10.9)
Average $I/\sigma(I)$	19.28 (1.97)	14.34 (3.41)	15.39 (1.83)	10.69 (2.57)
$R_{\text{meas}}(\%)^a$	7.3 (90.5)	8.6 73.7(55.3)	8.9 (89.4)	19.4 (83.0)
$CC^{1/2}(\%)$	99.9 (70.4)	99.8 (86.2)	99.9 (73.7)	99.5 (84.2)
Molecules/ asymmetric unit	1	1	1	2
Refinement				
$R_{\text{work}}^b/R_{\text{free}}^c(\%)$	18.19/18.91	16.57/18.39	19.87/23.54	18.39/21.98
No. of atoms	1955	1861	1732	15595
No. of residues	1600	1573	1580	15256
No. of water molecules	343	273	87	96
No. of Ligands	12	15	67	243
RMSD from ideality				
bond length (Å)	0.005	0.006	0.008	0.004
bond angle (°)	0.84	0.84	0.91	0.65
Ramachandran plot (%)				
Favoured	99.02	98.54	98.52	96.16
Allowed	0.98	1.46	1.48	3.74
Outliers	0.00	0.00	0.00	0.11
PDB accession code	8IN3	8IN4	8IN6	8IN1

The highest resolution shell is shown in parentheses.

^a $R_{\text{meas}} = \sum_{hkl} \{ N(hkl) / [N(hkl) - 1] \}^{1/2} \sum_i |I_i(hkl) - \langle I(hkl) \rangle| / \sum_{hkl} \sum_i I_i(hkl)$, where where

Table 1

X-ray data collection and structure-refinement statistics

$I_i(hkl)$ is the i th observation of the intensity of reflection hkl and $\langle I(hkl) \rangle$ is the mean over n observations.

$$^b R_{\text{work}} = \sum_{hkl} ||F_{\text{obs}}(hkl)| - |F_{\text{calc}}(hkl)|| / \sum_{hkl} |F_{\text{obs}}(hkl)|.$$

$^c R_{\text{free}}$ was calculated with an approximate 5% fraction of randomly selected reflections evaluated from refinement.

Table 1 (continued)

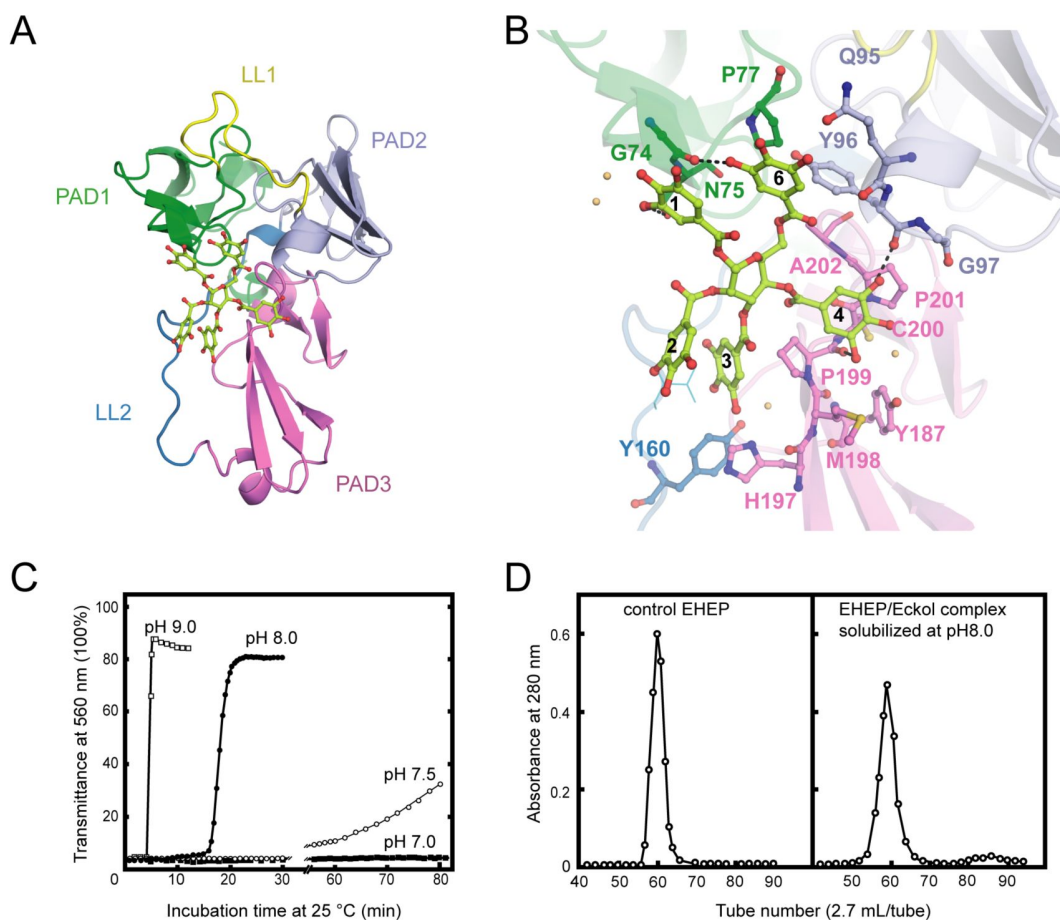


Figure 3.

Structure of EHEP-TNA. (a) The overall structure of EHEP-TNA. (A) the overall structure of EHEP-TNA. EHEP and TNA are shown by the cartoon and stick model, respectively. EHEP is colored as in Fig. 1. The C and O atoms of TNA are colored lemon and red, respectively. (B) Interaction of TNA (ball-stick in same color as (A)) with EHEP (cartoon in same color as (A)) in EHEP-TNA structure. The residues of EHEP in contact are labeled and shown by a ball-stick with N, O, and S atoms in blue, red, and brown, respectively. The C and O atoms of TNA are colored the same as (A), lemon and red, respectively. Dashed lines show hydrogen bonds. The water molecules stabilizing TNA was shown as light orange spheres. (C) Effect of pH on resolubilization of an EHEP-eckol precipitate. Buffers with pH 9.0, 8.0, 7.5, and 7.0 are presented as hollow square, solid circle, hollow circle, and solid square, respectively. (D) The EHEP-eckol precipitate was dissolved in 50 mM Tris-HCl (pH 8.0) and analyzed using a gel filtration column of Sephacryl S-100.

sheet contents of PAD2 and PAD3 (**Fig. S3B**). In the EHEP–TNA structure, the residues C93–Y96 of PAD2 folded into an α -helix and each β -sheet of the first β -strand in PAD3 elongated by incorporating one residue in the first (G¹⁷⁶) and second β -sheets (S¹⁸⁶) and three residues in the third β -sheet (H¹⁹⁷MP¹⁹⁹). The EHEP–TNA structure revealed that TNA binds to the center of the triangle formed by the three PADs, a positively charged surface (**Fig. 3A** and **Fig. S3C**). The binding pocket on EHEP surface was formed by the C-terminal α -helix of PAD1, the N-terminal α -helix of PAD2, and the middle part (loop) of PAD3 assisted by two long linker loops (LL1 and LL2). TNA was primarily bound to EHEP via hydrogen bonds and hydrophobic interactions (**Fig. 3B**). Gallic acid1, 4, and 6 interacted with EHEP via hydrogen bonds and additional hydrophobic contacts, whereas gallic acid2 and 3 only interacted hydrophobically with EHEP. The 3-hydroxyl groups of gallic acid1 and 6 individually formed a hydrogen bond with the main chain of G74 and the side chain of N75 in PAD1. The backbone carbonyl of Y96 and P199 in PAD2 and PAD3, respectively, formed a hydrogen bond with the 3,5-hydroxyl groups of gallic acid4. Additionally, some hydrogen bonds were formed between TNA and water molecules. TNA binding was also stabilized by hydrophobic interactions between the benzene rings of gallic acid and EHEP. For instance, gallic acid4 and 6 showed alkyl– π interaction with P201 and P77, respectively; moreover, gallic acid3 and 4 formed amide– π stackings with P199.

The EHEP–TNA structure clearly showed that TNA binds to EHEP without covalent bonds and the binding does not induce significant structural changes; thus, we attempted to recover EHEP from EHEP–TNA precipitates by adjusting the pH. As hypothesized, re-solubilization of the EHEP–phlorotannin precipitate is pH-dependent (**Fig. 3C**). The EHEP–TNA precipitate did not resolubilize at pH 7.0; however, after incubating for >1 h at pH 7.5, the precipitate started resolubilizing. Most of the precipitate rapidly resolubilized at an alkaline pH (≥ 8.0) after incubation for 15 min. Further, the resolubilized EHEP had the same elution profile as that of the natural EHEP (**Fig. 3D**) in SEC, suggesting that resolubilized EHEP maintained the native structure and its phlorotannin-precipitate activity (**Fig. S3D**).

Two domains of *akuBGL*

To reveal the structural basis of *akuBGL* recognition of laminarin and inhibition by TNA, we attempted to determine its structure. We soaked crystals in TNA as well as various substrate solutions but finally obtained the optimal resolution using crystal soaking in TNA. There was no electron density of TNA or something similar in the 2Fo–Fc and Fo–Fc map of the obtained structure; thus, we considered this structure as the apo form of *akuBGL*.

Two *akuBGL* molecules were observed in an asymmetric unit (MolA and MolB). These molecules lacked the N-terminal 25 residues (M1–D25), as confirmed using N-terminal sequencing analysis of purified natural *akuBGL*. This N-terminal fragment was predicted to be a signal peptide using the web server SignalP-5.0. The residues L26–P978 were constructed in both MolA and MolB with glycosylation, whereas the remaining C-terminal residues (A979–M994) could not be visualized as they were disordered. The electron density map of Fo–Fc revealed N-glycosylation at three residues, i.e., N113, N212, and N645 (**Figs. S4A, B**). N-glycosylation of GH enzymes prevents proteolysis and increases thermal stability^{28, 29}. Additionally, a study on β -glucosidase *Aspergillus terreus* BGL demonstrated that N-glycosylation of N224 affected the folding stability, even when it is located close to a catalytic residue³⁰. In *akuBGL*, all N-glycosylation sites were present on the surface, far away from the catalytic site. Therefore, we speculate that *akuBGL* glycosylation does not affect its activity. Except for the difference in visualized glycans resulting from glycosylation, MolA and MolB were similar, with a RMSD value of 0.182 for 899 Ca atoms; therefore, we used MolA for further descriptions and calculations.

The structure of *akuBGL* consisted of two independent GH1 domains, GH1D1 (L26–T494) and GH1D2 (D513–P978), linked by a long loop (D495–Y512) (**Fig. 4A**). There was little interaction between GH1D1 and GH1D2, only in a buried surface area comprising 2% of the total surface

(708.9 Å²) (Fig. S4C). GH1D1 and GH1D2 have a sequence identity of 40.47% and high structural similarity with an RMSD value of 0.59 Å for 371 Ca atoms (Fig. S5A upper panel).

Glucosidases of the GH1 family utilize the retaining mechanism with two glutamic acids for catalyzing glucoside hydrolysis. In general, the distance between the two catalytic oxygen atoms of the side chain of two glutamic acids is approximately 5 Å³¹. Sequence and structure alignment of GH1D1 and GH1D2 of *akuBGL* with other members of the GH1 family revealed that the second glutamate is conserved (E404), but the first glutamate is replaced by D192 in GH1D1. The oxygen atoms of the side chains between D192 and E404 of GH1D1 were 8.4 Å apart. In contrast, GH1D2 conserved two glutamic acids (E675 and E885) at the carboxyl termini of β -strands 4 and 7; the distance between oxygen atoms of E675 and E885 side chains was 5.1 Å (Fig. S5A bottom panel), similar to that of *Neotermes koshunensis* BGL (*NkBGL*, PDB ID 3VIH)³², *Nannochloropsis oceanica* BGL (*NoBGL*, PDB ID 5YJ7)³³, and *Spodoptera frugiperda* BGL (PDB ID 5CG0) (3.9–4.9 Å)³⁴. Furthermore, regarding the two other conserved essential regions for β -glucosidase activity, namely, glycone-binding site (GBS) and catalysis-related residues (CR), GH1D1 conserved neither GBS nor CR, whereas GH1D2 conserved both (Fig. 4B). Altogether, we suggest that GH1D1 does not possess catalytic activity. We expressed and purified the recombinant GH1D1, which did not show any hydrolytic activity toward O-PNG (Figs. S4D, E), although we could not rule out the effect of N-glycosylation.

Structural comparison of GH1D2 with other BGLs, including *NkBGL* (PDB ID 3VIH)³², rice (*Oryza sativa* L.) BGL (*OsBGL*, PDB ID 2RGL)³⁵, and microalgae *NoBGL* (PDB ID 5YJ7)³³, revealed the characteristics of each active pocket (Fig. S5B). *OsBGL* and *NoBGL* have a deep, narrow, and straight pocket, whereas GH1D2 and *NkBGL* have a broad and crooked pocket. Such active pocket shapes reflect the substrate preferences of *OsBGL* and *NoBGL*; they hydrolyze laminaribiose with no detectable activity toward laminaritetraose^{33,36}. Furthermore, the difference in the features of large active pockets between *NkBGL* and GH1D2, wherein GH1D2 often possesses an auxiliary site with several aromatic residues bound to the carbohydrate via CH– π interactions³⁷, may explain their substrate specificity. *NkBGL* efficiently hydrolyzes laminaribiose and cellobiose but has weak hydrolytic activity toward laminarin³⁸. In contrast, the GH1D2 of *akuBGL* has similar activity levels toward cellobiose and laminarin³⁹. Therefore, the GH1D2 of *akuBGL* may recognize larger substrates than that of other BGLs. Laminarin typically has a curved conformation; accordingly, narrow- and straight-shaped pockets are incompatible for binding. Furthermore, we docked GH1D2 with laminaritetraose, wherein the four glucose units formed extensive contacts with GH1D2. Hydrogen bonds involved the catalytic residues E675. In addition, several aromatic residues, such as W631, F677, W681, F689, Y819, Y846, W857, and W935, formed CH– π stacking (Fig. S6). Some interacting residues belonged to GBS and CR sites, such as E675, W631, Y819, and W935. Additionally, the docking structure revealed that the +3 and +4 glucose of laminaritetraose is located at the auxiliary binding site and that atom O1 of the +4 glucose is positioned outside the pocket (Fig. S6), implying that the auxiliary binding site with several aromatic residues (F677 and W681) of GH1D2 facilitates laminarin binding.

Inhibitor binding of *akuBGL*

As we could not obtain the complex structure of *akuBGL* with TNA, we performed docking calculations of *akuBGL* GH1D2 with TNA to explore the inhibition mechanism. The docking model of *akuBGL*–TNA showed that seven gallic acid rings of TNA formed an extensive hydrogen bond network with *akuBGL* in the binding pocket (Fig. 5). The hydroxyl groups of TNA formed hydrogen bonds with the residues N552, E675, D735, K739, K759, Q840, T844, D852, and K859 of GH1D2. Moreover, benzene rings showed hydrophobic interactions with several hydrophobic residues. In particular, stable π – π stacking was observed between TNA and residues F547, W631, F689, Y846, W857, and W935. Among these residues, the conserved E675 was the catalytic residue, and W631, W935, and E934 contributed to GBS and CR sites.

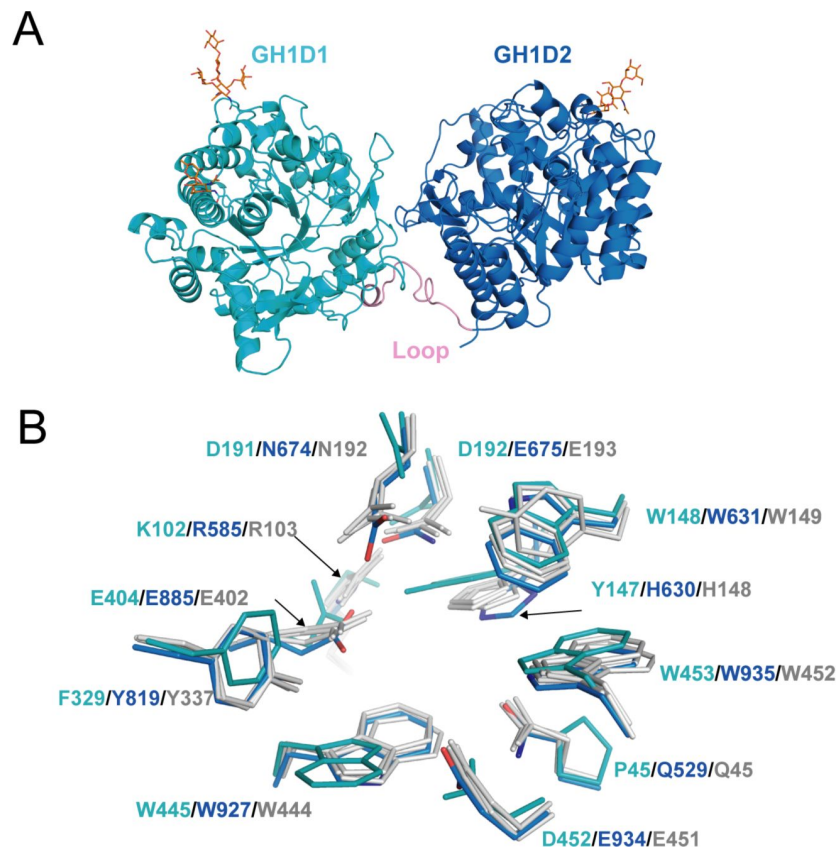


Figure 4.

Structure of *akuBGL*. (A) Overall structure. The GH1D1 (light blue) and GH1D2 (cyan) domains are linked by a long loop (linker-loop) colored in pink. The N-linked glycans were shown in the orange stick. (B) Residues superposition of the GBS and CR sites of the domains GH1D1 (cyan), GH1D2 (light blue) with β -glucosidase structures from termite *Neotermes koshunensis* (*NkBGL*, grey(PDB ID 3VIH) ³², β -glucosidase from rice (*OsBGL*, gray, PDB ID 2RGL) ³⁵, β -glucosidase from *Bacillus circulans* sp. *Alkalophilus* (grey, PDB ID 1QOX) ⁶⁷. Only the residues numbers of GH1D1 (cyan), GH1D2 (light blue), and *NkBGL* (grey) are shown for clarity.

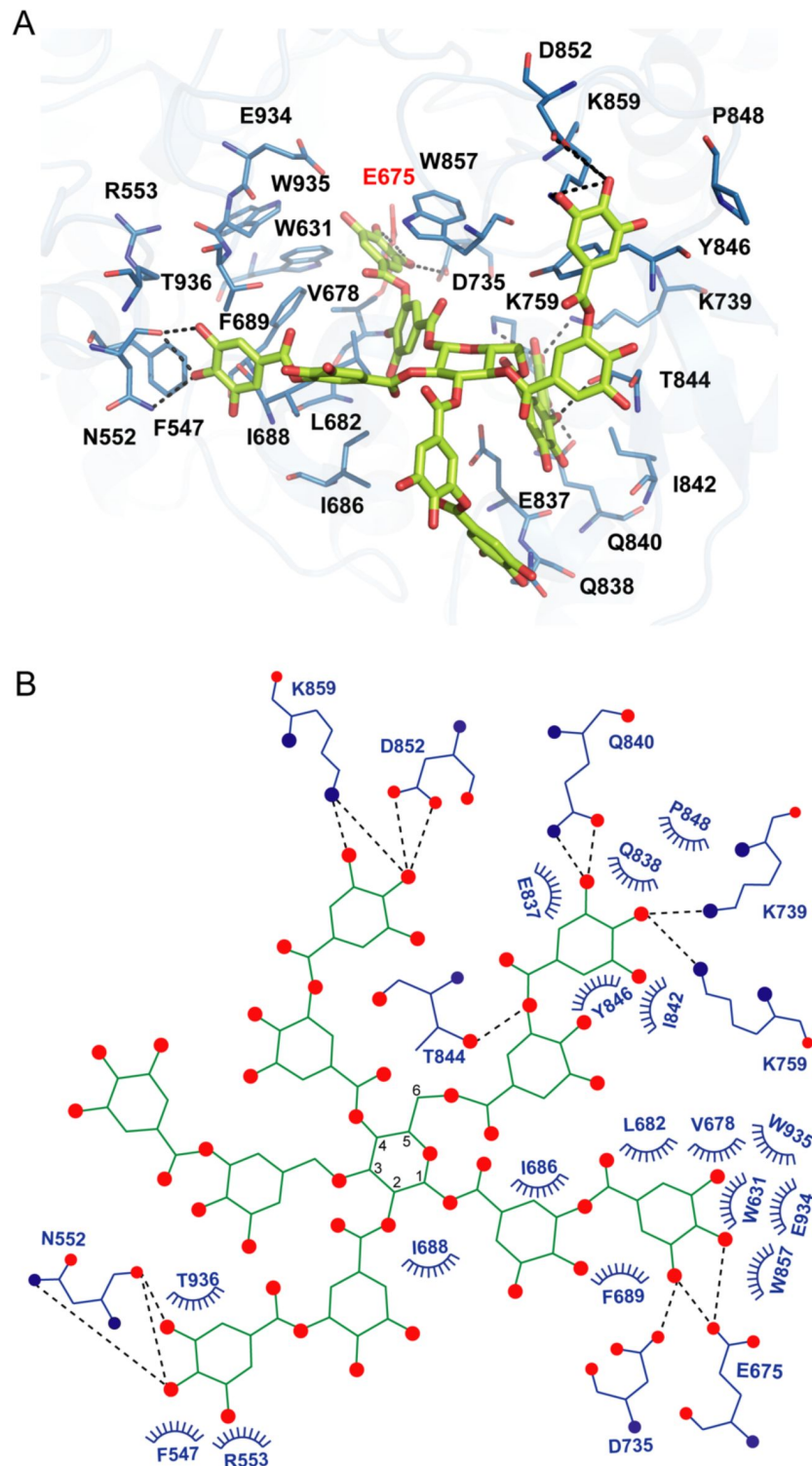


Figure 5.

Docking model of *akuBGL* with TNA. (A) Detailed interaction between *akuBGL* and TNA in the docking model. TNA is shown in the green stick model. The hydrogen bonds are shown as dashed lines. (B) A 2D diagram of the interaction between *akuBGL* and TNA shown in (A). The hydrogen bonds are shown as dash lines, and the hydrophobic contacts as circular arcs.

In addition, we performed a docking calculation of GH1D2 with the characteristic inhibitors eckol and phloroglucinol⁴⁰. The binding mechanisms of eckol and phloroglucinol were similar to those of TNA but with different contact residues (**Fig. S7**). For eckol, the six hydroxyl groups formed hydrogen bonds with residues E675, D735, E737, K759, E885, and E934. Additionally, residues W631, F677, F689, Y819, W857, W935, F943, and W927 formed π - π stacking with eckol. For phloroglucinol, the three hydroxyl groups formed hydrogen bonds with E675, E885, and E934, whereas residues W631, F689, Y819, W857, W935, and W927 formed π - π stacking with the benzene ring.

The docking scores of the inhibitors TNA, eckol, and phloroglucinol were -8.8, -7.3, and -5.7 kcal/mol, respectively, whereas the substrate laminaritetraose had a docking score of -6.6 kcal/mol. Those docking scores corroborated well with the inhibition activity toward *akuBGL*, that TNA had a more robust inhibition activity than phloroglucinol⁴, indicating that the docking results are reasonable. In summary, the three inhibitors interacted with *akuBGL* through similar binding mechanisms to occupy the substrate-binding site, suggesting a reversible competitive inhibition mechanism.

Discussion

In marine habitats, the ecological interactions between brown algae and herbivores dominate marine ecosystems⁴¹. The *akuBGL*-phlorotannin/laminarin-EHEP system represents the feeding defense-offense associations between *A. kurodai* and brown algae. We focused on this system to understand the molecular mechanism at the atomic level. In contrast to most GH1 BGLs containing one catalytic GH1 domain, *akuBGL* consists of noncatalytic GH1D1 and catalytic GH1D2. The noncatalytic GH1D1 may act as a chaperone of GH1D2, as we successfully overexpressed GH1D1 but failed to do the same for GH1D2. Such multi-GH1D assembly and a similar function has been suggested in β -glucosidase *CjCEL1A* of *Corbicula japonica*⁴² and glycosidase *LpMDGH1* of the shipworm *Lyrodus pedicellatus*⁴³. *CjCEL1A* has two tandem GH1Ds with a sequence identity of 43.41%⁴². Two catalytic glutamic acids and the residues related to substrate binding are conserved in the second GH1D, whereas the first GH1 domain lacks these conserved residues and may play a role in folding the catalytic domain. *LpMDGH1* consists of six GH1Ds, among which GH1D2, 4, 5, and 6 contain the conserved residues for activity, whereas others do not contain these residues and might be involved in protein folding or substrate interactions⁴³. Future assay of GH1D2 and inactive mutants is the complement to validate the molecular mechanism of *akuBGL*.

BGLs have different substrate preferences in the degree of polymerization and type of glycosidic bond. In general, BGLs prefer to react with mono-oligo sugars over polysaccharides. For instance, *OsBGL*, *NoBGL*, and *NkBGL* hydrolyze disaccharides (cellobiose and laminaribiose) but display no or weak activity toward polysaccharides (cellulose and laminarin)^{33, 36, 38}. The structure of GH1D2 explained the substrate preference for the polysaccharide laminarin. GH1D2 contains an additional auxiliary site composed of aromatic residues (**Fig. S5B**) in the substrate entrance pocket, which putatively enables it to accommodate a long substrate, contributing to *akuBGL* activity toward laminarin, as supported by docking calculations (**Fig. S6**). In addition, docking analysis of *akuBGL* GH1D2 with inhibitors (TNA, phlorotannin, eckol, and phloroglucinol) revealed that these inhibitors bind to the substrate-binding site via hydrogen bonds and hydrophobic interactions similar to laminarin. Such binding mechanisms suggest the presence of competitive inhibition to occupy the binding site, consistent with previous research⁴. Future kinetic experiments are required to validate quantitatively the competitive inhibition of phlorotannin against *akuBGL*.

EHEP, expressed in the midgut of *A. kurodai*, was identified as an antidefense protein, protecting the hydrolysis activity of *akuBGL* from phlorotannin inhibition ⁴⁴. Such an ecological balance also exists between plants and their predatory mammals and insects. Similar to brown algae, plants use the toxic secondary metabolite tannins as their defense mechanism against predators, which constitute 5%–10% of dry weight of leaves. In vertebrate herbivores, tannins reduce protein digestion ⁴⁴. In phytophagous insects, tannins may be oxidized at the alkaline pH of the insect midgut and cause damage to cells ⁴⁵. The evolution of plant–herbivore survival competition has led to the development of remarkably unique adaptation strategies. Mammals feeding on plants that contain tannin may overcome this defense by producing tannin-binding proteins, proline-rich proteins, and histatins ⁴⁶ ⁴⁷. Proline constitutes at least 20% of the total amino acid content in proline-rich proteins; for some species, the proportion of proline reaches 40%. Histidine constitutes 25% of total amino acid content in histatins. Both proline-rich proteins and histatins are unfolding proteins with randomly coils in solution. In caterpillars, the oxidation damage of tannin is reduced by the low oxygen level. Some insects use the peritrophic membrane to transport tannins into the hemolymph, where they are excreted ⁴⁸. Additionally, the peritrophic envelop protects insects from tannins by forming an impermeable barrier to tannins ⁴⁴. *A. kurodai* uses a similar strategy to mammals by secreting the tannin-binding protein EHEP. Although EHEP has a completely different amino acid composition with proline-rich proteins and histatins, EHEP also binds to phlorotannin. Therefore, EHEP may be a specific counteradaptation that allows *A. kurodai* to feed on brown algae, as there are no homologous proteins in other organisms.

The three PADs of EHEP are arranged in a triangle shape, forming a large cavity on the surface at the triangle center to provide a ligand-binding site. EHEP has a positively charged surface at a pH of <6.0, whereas the surface becomes negatively charged at a pH of >7.0 (**Fig. S3C**). Meanwhile, TNA has a pKa of 4.9–8 ⁴⁹ ⁵¹, showing minor negative charges at an acidic pH and the highest negative charge at a pH of >7.0 ⁵². Therefore, TNA binds to EHEP at a pH of <6.0 (pH of crystallization = 4.5), but it shows charge repulsion with EHEP at a pH of >8.0. Altogether, TNA is protonated and behaves as a hydrogen bond donor when the pH is below its pKa, whereas when the pH is above its pKa, TNA is deprotonated and the hydrogen bonding cannot be maintained. As losing hydrogen bonds and increasing repulsive forces at a pH >8.0, the precipitated EHEP–TNA could dissolve in the buffer of pH >8.0. This pH-induced reversible interaction also occurred in other proteins, such as BSA, pepsin, and cytochrome C ⁵³. The phlorotannin members share a similar structure with TNA; thus, we speculate that the EHEP–phlorotannin complex also exhibits a pH-induced reversible interaction. *In vivo*, the pH of the digestive fluid of *A. kurodai* is approximately 5.5 ⁴, which favors the binding of EHEP to phlorotannin. In the alkaline hindgut ⁵⁴, the EHEP–phlorotannin disassociates (**Fig. 6**), and the phlorotannin is subsequently excreted from the anus.

Based on the EHEP–TNA structure and docking models of *akuBGL*–inhibitor/substrate, we proposed a mechanism of phlorotannin inhibition on *akuBGL* activity and EHEP protection from phlorotannin inhibition (**Fig. 6**). Because laminarin lacks the benzene rings essentially to form CH– π stacking interactions with EHEP, the EHEP can be considered not binding with the laminarin. In the absence of EHEP, phlorotannin occupies the substrate-binding site of *akuBGL*, inhibiting the substrate from entering the active site and resulting in no glucose production. In the presence of EHEP, it competitively binds to phlorotannin, freeing the *akuBGL* pocket. Then, the substrate can enter the active pocket of *akuBGL* and glucose can be produced normally. The digestive fluid of *A. kurodai* contains EHEP at a high concentration (>4.4 μ M) ⁴, which is slightly higher than the concentration of EHEP (3.36 μ M) that protects *akuBGL* activity (**Fig. 1B**). The high concentration of EHEP allows *A. kurodai* feeding of phlorotannin-rich brown algae. The balance between phlorotannin inhibition and protection is controlled by the concentrations of phlorotannin and EHEP *in vivo*.

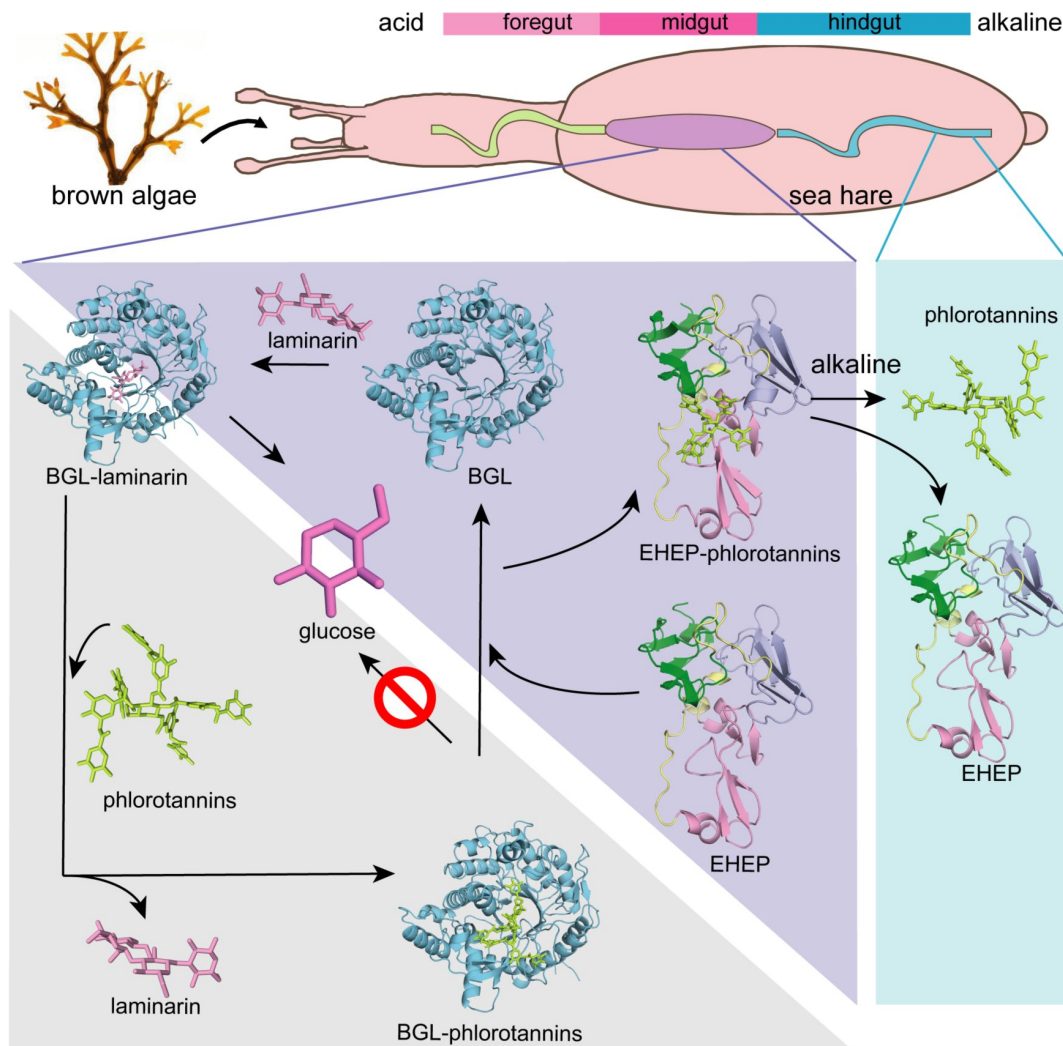


Figure 6.

Proposed molecular mechanisms. Proposed molecular mechanisms of TNA inhibition of *akuBGL* activity and EHEP's protective effects of *akuBGL* in *akuBGL*-phlorotannin/laminarin-EHEP system (light purple triangle). The digestive tract of *A. kurodai* consists of foregut (blue), midgut (purple), and hindgut (blue) (top). The bar chart above depicts the pH of the digestive tract, with pink denoting acid and blue denoting alkalinity.

The *akuBGL*–phlorotannin/laminarin–EHEP system is the digestive–defensive– offensive associations between algae and herbivores. Our study presented the molecular mechanism of this system at the atomic level, providing a molecular explanation for how the sea hare *A. kurodai* utilizes EHEP to protect *akuBGL* activity from phlorotannin inhibition. Further, such a feeding strategy has attracted attention for producing glucose as a renewable biofuel source, so our studies provide a molecular basis for the biofuel industry applications of brown algae.

Materials and Methods

EHEP and *akuBGL* preparation

Natural EHEP (22.5 kDa) and *akuBGL* (110 kDa) were purified from *A. kurodai* digestive fluid as described previously⁴. For crystallization, we further added one step for purification of EHEP using size exclusion chromatography (HiLoad 16/60 Superdex 75, GE Healthcare), for which the column was equilibrated with 20 mM MES-NaOH buffer (pH 6.5). Obtained EHEP was then concentrated to 15–25 mg/mL using Vivaspin-4 10K columns (Sartorius, Göttingen, Germany). About *akuBGL*, we exchanged buffer from 20 mM Tris-HCl pH 7.0 to 20 mM Bis-tris pH 6.0 and concentrated it to 11 mg/mL using Amicon with a cutoff of 50 kDa.

To verify whether the chemical modification which was indicated by previous study²² affects the function of EHEP, we prepared recombinant EHEP (recomEHEP) without the N-terminal signal peptide (1–20 aa) and chemical modifications²². EHEP cDNA was obtained via reverse transcription–polymerase chain reaction (RT–PCR) using the total RNA of *A. kurodai* as a template. The reamplified fragment was digested and ligated to a plasmid derived from pET28a (Novagene, Darmstadt, Germany). We transformed the plasmid containing recomEHEP into *E. coli* B834(DE3) pARE2 and expressed it with a C-terminal hexahistidine-tag. The cells were cultured in lysogeny broth (LB) medium with the antibiotics kanamycin (25 mg/L) and chloramphenicol (34 mg/L) until the optical density at 600 nm (OD₆₀₀) reached 0.6. Subsequently, overexpression was induced by adding 0.5 mM isopropyl b-D-L-thiogalactopyranoside for 20 h at 20 °C. After harvesting by centrifugation, the cells were resuspended in a buffer containing 50 mM Tris-HCl pH 7.4, 300 mM NaCl, DNase, and lysozyme and were disrupted via sonication. The insoluble part was removed by centrifugation for 30 min at 40000 ×g at 4 °C. We loaded the supernatant onto a 5-mL HisTrap HP column and the recomEHEP was eluted using increasing concentrations of imidazole (0–500 mM). The purified proteins were dialyzed against a solution containing 50 mM Tris-HCl pH 7.4 and 50 mM NaCl and subsequently loaded onto a HisTrap Q column and eluted by a linear gradient of a solution containing 50 mM Tris-HCl and 1 M NaCl. Fractions containing recomEHEP were concentrated and then purified using a gel filtration column (HiLoad 16/60 superdex 75 pg) equilibrated with 20 mM sodium acetate pH 6.0 and 100 mM NaCl. We collected the fractions containing recomEHEP and concentrated them to 2.1 mg/mL using Amicon (Merck, American).

DNA encoding *akuBGL* without signal peptide was codon optimized (GENEWIZ, China) for overexpress in *E. coli*, and cDNA encoding GH1D1 domain was subcloned to a modified pET-32a vector (Invitrogen, USA) with an N-terminal TrxA tag and a 6×His tag followed by a TEV protease recognition site. The plasmid pET-32a-GH1D1 was electorally transformed into *E.coli* origami2 cells. We grew the cells in LB medium supplemented with 100 µg/mL of ampicillin at 37 °C until the OD₆₀₀ reached 0.6. Then the cultures were cooled and induced with 0.5 mM isopropyl-β-D-thiogalactopyranoside at 20 °C for 16 h. After resuspending the harvested cells in a buffer containing 50 mM Tris-HCl (pH 7.4), 250 mM NaCl, and 5% glycerol, we disrupted the cells by sonication. The cell lysate was centrifugated at 40,000 ×g for 30 min at 4 °C. The supernatant was filtered by 0.45 µm membrane and then loaded onto a HisTrap HP column (GE, USA). After washing with lysis buffer supplemented with 20 mM imidazole, the recomGH1D1 was eluted linearly from the column with lysis buffer supplemented with 500 mM imidazole. The fractions containing recomGH1D1 were added with TEV protease at a ratio of 1:10 and dialyzed against the buffer

containing 50 mM Tris (pH 7.4), 50 mM NaCl, and 2 mM DTT at 4 °C overnight. Then, we purified the recomGH1D1 by a Histrap HP column again, and collected the flowthrough. Finally, we purified the recomGH1D1 by size exclusion chromatography on a Hiload 16/60 200 pg column (GE, USA) equilibrated with a buffer containing 20 mM Bis-Tris (pH 6.0). The recomGH1D1 was concentrated to 1.5 mg/mL by centrifugation using Amicon Ultra with 10 kDa. cutoff.

N-terminal sequencing of *akuBGL*

We performed a N-terminal sequencing of purified *akuBGL* using the Edman degradation method. *akuBGL* was separated by SDS–PAGE, followed by electrophoretic transfer onto a PVDF (polyvinylidene fluoride) membrane (GE, American). The membrane was subsequently stained with Ponceau S solution. The band corresponding to *akuBGL* was excised and analyzed using a PPSQ-53A Protein sequencer (Shimadzu, Japan) at the Instrumental Analysis Service of Hokkaido University.

Effects of TNA on *akuBGL* activity with or without EHEP

Due to the yellow color of TNA, which affects absorbance at 420 nm of the reaction product o-nitrophenol of *akuBGL*, we used HPLC to measure the *akuBGL* activity in the reaction system involving TNA. Ortho-nitrophenyl-β-galactoside (ONPG) was used as a substrate to measure *akuBGL* activity. The reaction system (100 μL) included 2.5 mM ONPG, 49 nM *akuBGL*, and different TNA concentrations in a reaction buffer (50 mM CH₃COONa pH 5.5, 100 mM NaCl, and 10 mM CaCl₂). After incubation for 10 min at 37 °C, 100 μL of methanol was added to each sample to terminate the reaction. Then, the mixture was centrifuged for 10 min at 15000 ×g at 4 °C and the supernatant was used for analyzing *akuBGL* activity via HPLC. To measure the protective effect of EHEP on *akuBGL*, we added different amounts of EHEP to the reaction system.

RecomGH1D1 activity assay

We measured simply the activity of the recomGH1D1 using a spectrophotometer because the reaction product o-nitrophenol is yellow. The reaction system (100 μL) included 2.5 mM ONPG, 49 nM *akuBGL* or recomGH1D1 in a reaction buffer (50 mM CH₃COONa pH 5.5, 100 mM NaCl, and 10 mM CaCl₂). After incubation for 10 min at 37 °C, the reaction was terminated by adding 100 μL of 500 mM Na₂CO₃. The absorbance at 420 nm was measured using a SpectraMax spectrophotometer (Molecular Devices, Japan).

Binding assay for recomEHEP with TNA

We measured the binding activity of recomEHEP using precipitation analysis in same method of natural EHEP, as described previously [42](#). Briefly, recomEHEP or EHEP was incubated with TNA at 25 °C for 90 min and centrifuged for 10 min at 12000 ×g at 4 °C. Then, we washed the precipitates twice and resuspended them in an SDS–PAGE loading buffer for binding analysis.

Resolubilization of the EHEP–eckol precipitate

A mixture of 2 mg of EHEP and 0.4 mg of eckol was incubated at 37 °C for 1 h, followed by centrifugation at 12000 ×g for 10 min, and the supernatant was removed. The sediment was dissolved in 50 mM Tris-HCl buffer at different pH (7.0–9.0) and the absorbance at 560 nm was measured over time. After checking the elution peak by SDS–PAGE, the resolubilized EHEP was analyzed by a Sephacryl S-100 HR column (2.0 ×110 cm). Moreover, the eckol/dieckol-binding activity of resolubilized EHEP was assessed, as mentioned above in this section.

Crystallization and data collection

The crystallization, data collection, and initial phase determination of EHEP were described previously [22](#). As EHEP precipitates when bound to TNA, we could not cocrystallize EHEP with TNA. Therefore, we used the soaking method to obtain the EHEP–TNA complex. Owing to the poor

reproducibility of EHEP crystallization, we used a co-cage-1 nucleant ⁵⁵ to prepare EHEP crystals for forming the complex with TNA. Finally, we obtained high-quality EHEP crystals under the reservoir solution containing 1.0 M sodium acetate, 0.1 M imidazole (pH 6.5) with co-cage-1 nucleant. Subsequently, we soaked the EHEP crystals in a reservoir solution containing 10 mM TNA at 37 °C for 2 days; then, they were maintained at 20 °C for 2 weeks. Next, we soaked the EHEP crystals in a reservoir solution containing 10 mM phloroglucinol. For data collection, the crystal was soaked in a cryoprotectant solution containing 20% (v/v) glycerol along with the reservoir solution. Diffraction data were collected under a cold nitrogen gas stream at 100 K using Photon Factory BL-17 (Tsukuba, Japan) or Spring 8 BL-41XU (Hyogo, Japan).

For *akuBGL* crystallization, the initial crystallization screening was performed using the sitting-drop vapor-diffusion method with Screen classics and Classics II crystallization kits (Qiagen, Hilden, Germany) and PACT kits (Molecular Dimensions, Anatrache, Inc.) at 20 °C. Crystallization drops were set up by mixing 0.5 µL of protein solution with an equal volume of the reservoir solution. The initial crystals were obtained under condition no. 41 (0.1 M sodium acetate pH 4.5 and 25% polyethylene glycol [PEG] 3350) of Classics II, no. 13 (0.1 M MIB buffer [25 mM sodium malonate dibasic monohydrate, 37.5 mM imidazole, and 37.5 mM boric acid], with pH 4.0 and 25% PEG 1500), and no. 37 (0.1 M MMT buffer [20 mM DL-malic acid, 40 mM MES monohydrate, and 40 mM Tris], with pH 4.0 and 25% PEG 1500) of PACT. After optimization by varying the buffer pH and precipitant concentration and adding co-cage1 nucleant, the optimal crystals were obtained using 0.1 M sodium acetate pH 4.5, and 20% PEG 3350 with a co-cage1 nucleant at a protein concentration of 5.4 mg/mL. Diffraction data were collected under a cold nitrogen gas stream at 100 K using Photon Factory BL-1A (Tsukuba, Japan) after cryoprotection by adding glycerol to a 20% final concentration into the reservoir solution. The optimal resolution of diffraction data was obtained by soaking a crystal with 5 mM TNA in the reservoir buffer at 37 °C for 4 h.

All datasets were indexed, integrated, scaled, and merged using *XDS/XSCALE* program ⁵⁶. Statistical data collection and process are summarized in **Table 1**.

Structure determination and refinement

For EHEP structure determination, after initial phasing via the native-SAD method ^{22, 57}, the model was obtained and refined with *auto-building* using *Phenix.autobuild* of *Phenix* software suite ⁵⁸. The obtained native-SAD structure was used as a model for rigid body refinement using *phenix.refine* ⁵⁹ of *Phenix* software suite with a native data at high resolution of 1.15 Å. The structure of EHEP was automatically rebuilt using *Phenix.autobuild* of the *Phenix* software suite again ⁵⁸. Several rounds of refinement were performed using *Phenix.refine* of the *Phenix* software suite ⁵⁸, alternating with manual fitting and rebuilding using *COOT* program ⁶⁰. The final refinement statistics and geometry are shown in **Table 1**.

The structure of the EHEP–TNA complex was determined via the molecular replacement (MR) method using the EHEP structure as a search model with *Phaser* of *Phenix* software suite ⁶¹. The electron density block of TNA was clearly shown in both $2F_o - F_c$ and $F_o - F_c$ maps. Subsequently, TNA structure was manually constructed, followed by several rounds of refinement using *Phenix.refine* ⁵⁸, with manual fitting and rebuilding using *COOT* ⁶⁰. We also determined the structure of phloroglucinol-soaked crystals at a resolution of 1.4 Å via the MR method using the refined EHEP structure as a search model with *Phaser*, but no electron density block of phloroglucinol was obtained. Therefore, we referred to this structure as the apo form (apo structure2). The final refinement statistics and geometry are shown in **Table 1**.

We determined the structure of *akuBGL* via the MR method using *Phaser* of *Phenix* software suite ⁶¹. We used one GH domain (86–505 aa) of β -klotho (PDB entry: 5VAN) ⁶² as the search model. This GH domain of β -klotho shares 30% sequence identity with *akuBGL*. Four GH domains of two

molecules in an asymmetric unit were found and subsequently rebuilt with *Phenix_autobuild* of *Phenix* software suite ⁵⁸[\[58\]](#). Finally, refinement of *akuBGL* structure was performed as described for EHEP.

Docking studies of *akuBGL* with phlorotanins and laminarins

We used Schrodinger Maestro program for performing docking studies ⁶³[\[63\]](#). First, we superimposed the structure of *OsBGL* mutant complexed with cellotetraose (PDB ID 4QLK ⁶⁴[\[64\]](#)) to that of *akuBGL* GH1D2 to define the ligand position in the ligand-binding cavity. Then, we modified the structure of the *akuBGL* GH1D2 using wizard module to remove water molecules and add hydrogen atoms for docking. The 2D structures of the inhibition ligands, including TNA, phloroglucinol, and eckol, were downloaded from Pubchem ⁶⁵[\[65\]](#) and further converted to 3D structures using the LigPrep module of Schrodinger Maestro program. The structure of the substrate laminaritetraose was extracted from the *Zobellia galactanivorans* β -glucanase–laminaritetraose complex structure (PDB ID: 4BOW) ⁶⁶[\[66\]](#). Then, a receptor grid was constructed in the center of the ligand-binding cavity. We performed docking using the Glide standard precision mode without any constraints. The optimal binding pose was determined using the lowest Glide score, and docked structures were analyzed using PyMol.

Data Availability

The atomic coordinates were deposited in the PDB with the accession codes as follows: EHEP with 1.15 Å resolution (8IN3), EHEP with 1.4 Å resolution (8IN4), EHEP complexed with tannic acid (8IN6), *akuBGL*(8IN1).

Acknowledgements

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Supporting information

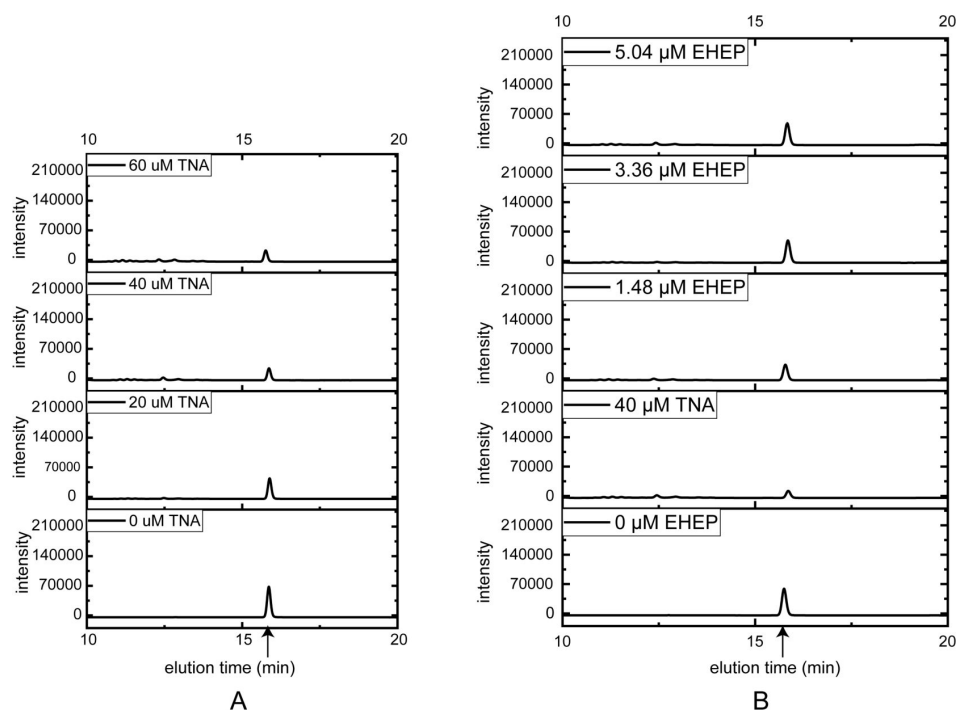


Fig. S1.

HPLC profiles of *akuBGL* activity toward ortho-Nitrophenyl-β-galactoside. (A) TNA inhibition *akuBGL* by increasing the concentration of TNA with 0.049 μM *akuBGL* and 2.5 mM ortho-Nitrophenyl-β-galactoside. (B) EHEP protects *akuBGL* from TNA inhibition by increasing the concentration of EHEP. The product peak was marked by a black arrow.

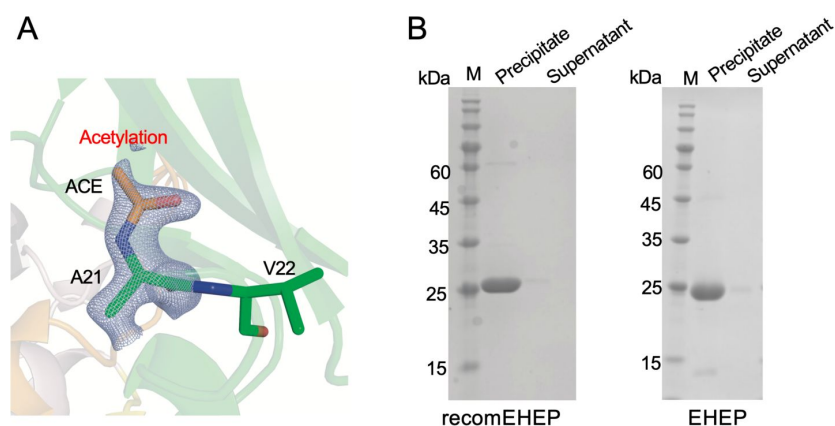


Figure. S2.

(A) Acetylation modification on the N-terminal residue A21. The structure is shown as sticks with an omitted map of the acetylation of A21 at a 3.3 σ level (blue-white). (B) TNA binding activity of recomEHEP and EHEP. SDS-PAGE was run using a mixture of recomEHEP/EHEP and TNA.

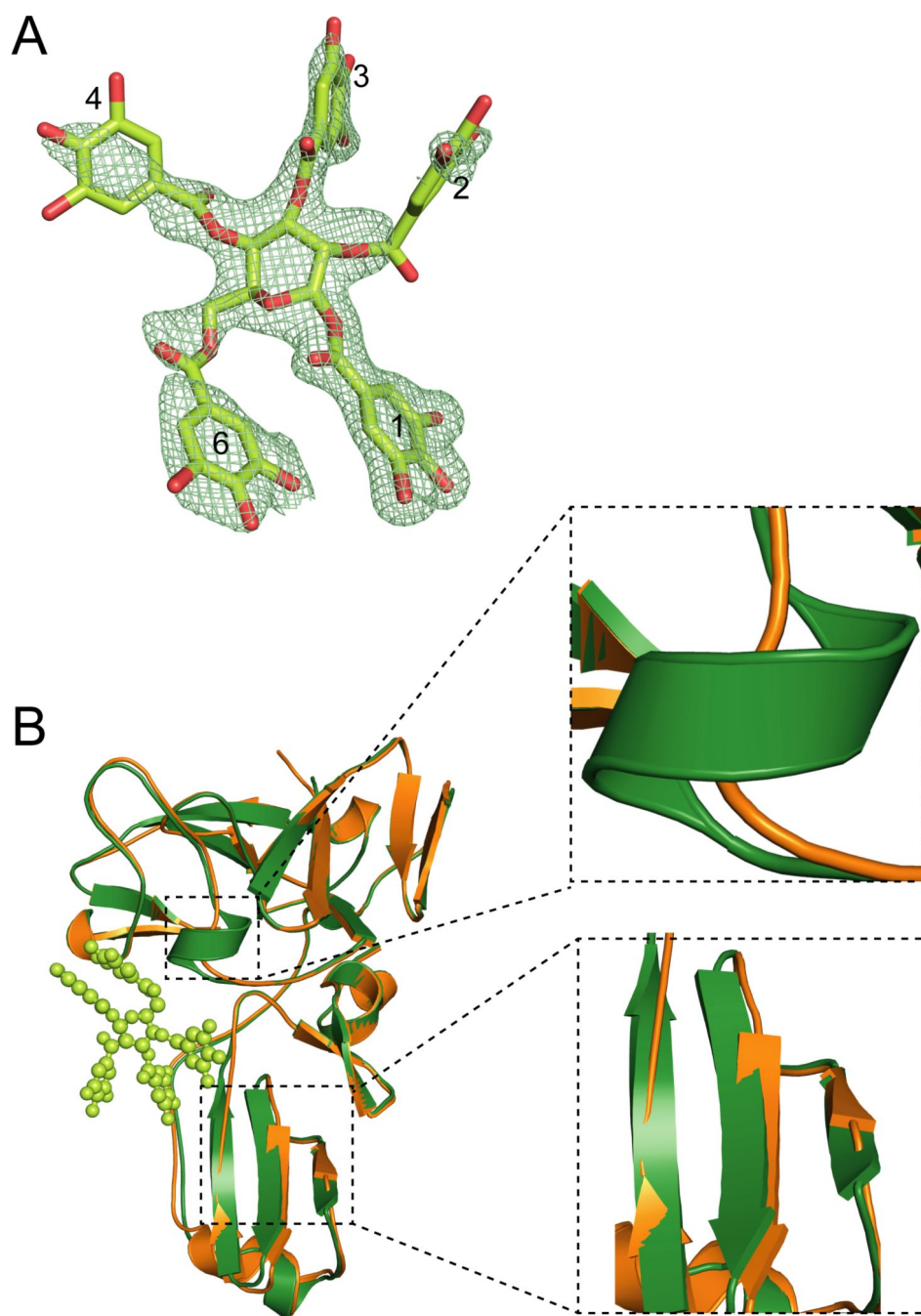
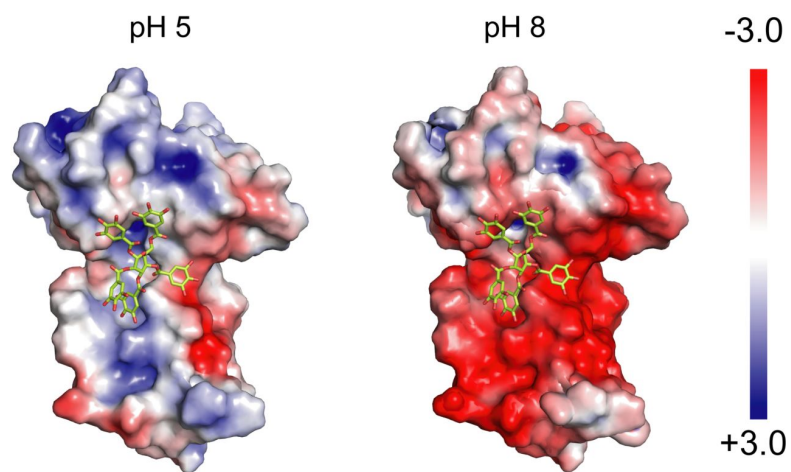


Figure S3.

EHEP-TNA structure and structural comparisons. (A) TNA structure (stick model) in EHEP-TNA with an omitted map countered at the 2.0 σ level. The O and C atoms are colored red and lemon, respectively. (B) Structure superposition of apo EHEP (green) with EHEP-TNA (orange). The conformational changes are marked by black dotted boxes and zoomed on the right. (C) Electrostatic potential of the EHEP surface in the EHEP-TNA complex at different pH values, which were calculated using APBS-PDB2PQR software suite. (D) Activity of the resolubilized EHEP. The EHEP-eckol precipitate was resolubilized in Tris-HCl (pH 8.0) buffer and then the supernatant was purified using a Sephacryl S-100 HR column (2.0 $\times$ 110 cm) to measure eckol/dieckol-binding activity.

C



D

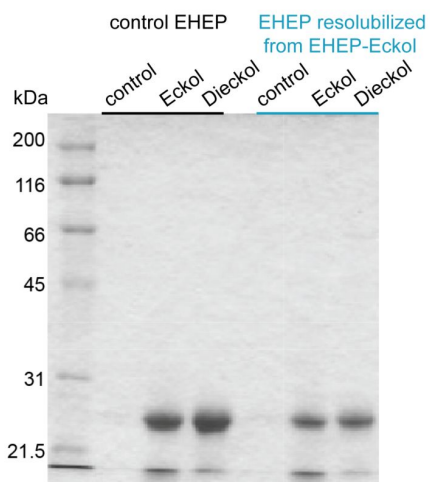


Figure S3. (continued)

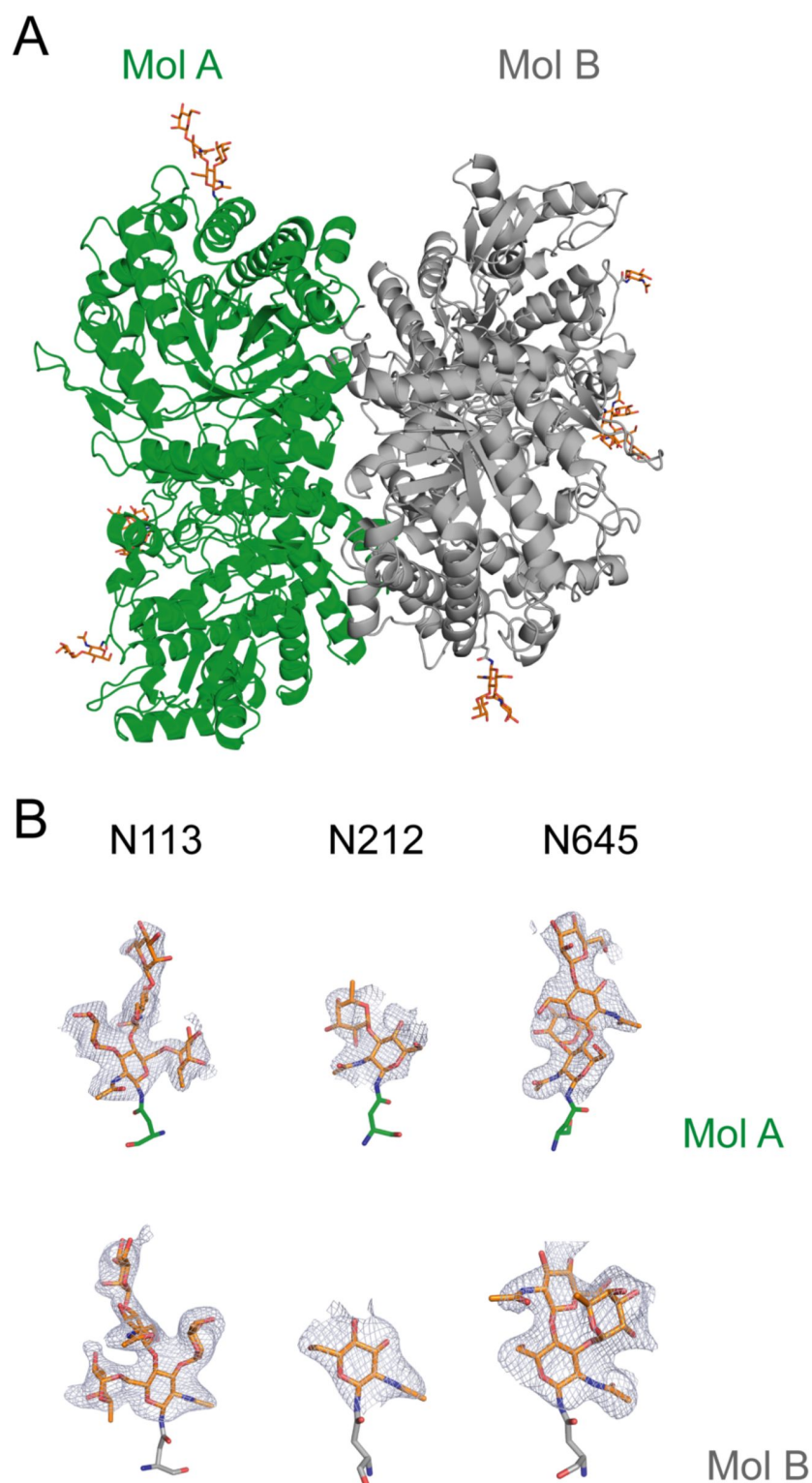


Figure S4.

akuBGL structure. (A) Two *akuBGL* molecules in the asymmetric unit, colored green and grey, respectively. The glycosylation sites are shown as orange sticks, with O and N atoms in red and blue, respectively. (B) The glycosylation chains with omitted density maps are countered at the 2.0 σ level. (C) The interface between the GH1D1 (cyan) and GH1D2 (blue). The key interactions between the two domains are shown as black dashed lines. (D) Size exclusion chromatogram of the purified GH1D1 domain. The inset shows the SDS-PAGE analysis of the GH1D1 domain. (E) Galactoside hydrolytic activity of the recomGH1D1 toward ortho-Nitrophenyl- β -galactoside. The average and standard deviation of the relative activity were estimated from three independent replicates (N = 3).

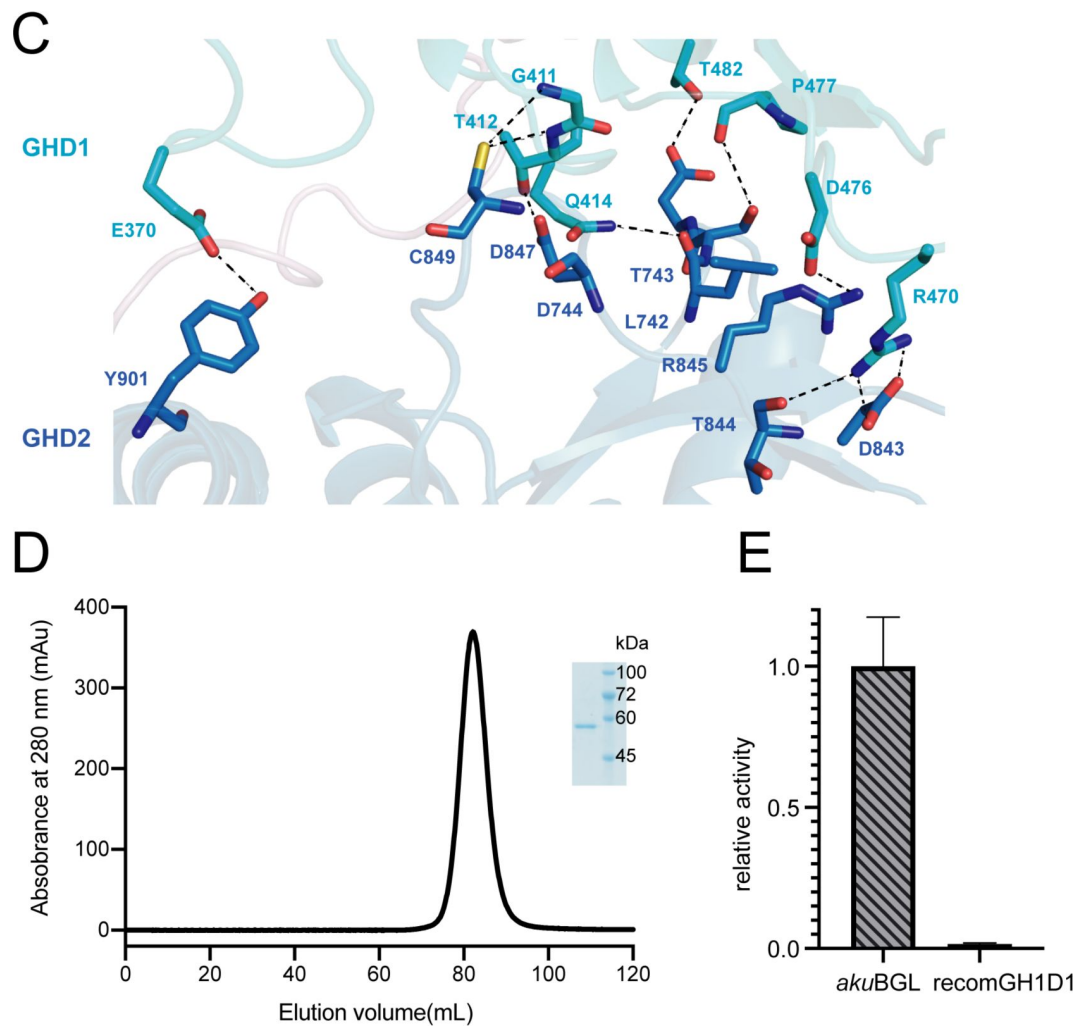


Figure S4. (continued)

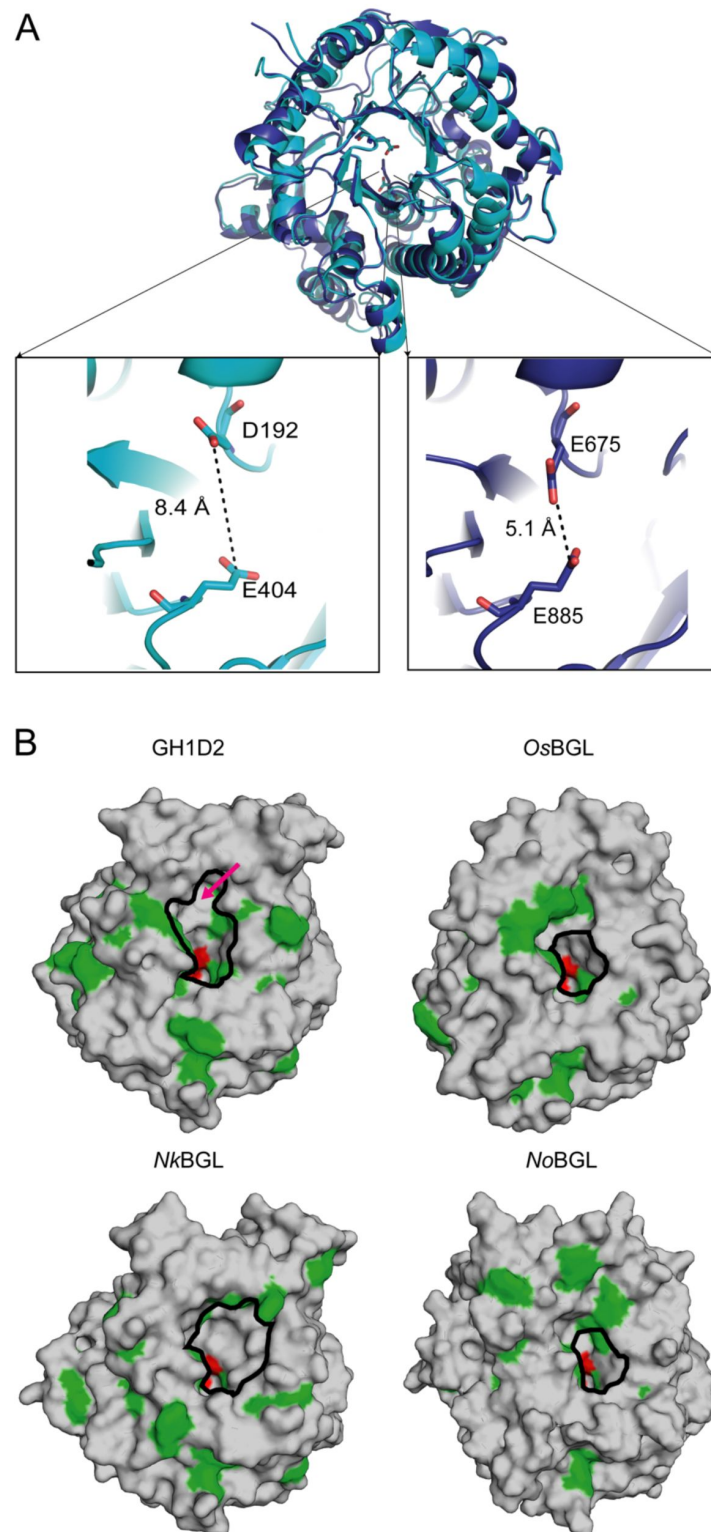


Figure S5.

Structural comparison of BGLs. (A) Structure superposition of GH1D1(cyan) and GH1D2 (blue). The enlarged picture shows the distance of conceivable catalytic residues in GH1D1 and GH1D2. (B) Surface representations of GH1D2, *NkBGL*(PDB ID 3VIH) ³², *OsBGL*(PDB ID 2RGL) ³⁵, and *NoBGL* (PDB ID 5YJ7) ³³ with the aromatic and catalytic residues colored green and red, respectively. The red arrow indicates the location of the auxiliary site of GH1D2. The active pockets are highlighted by a black circle on each surface.

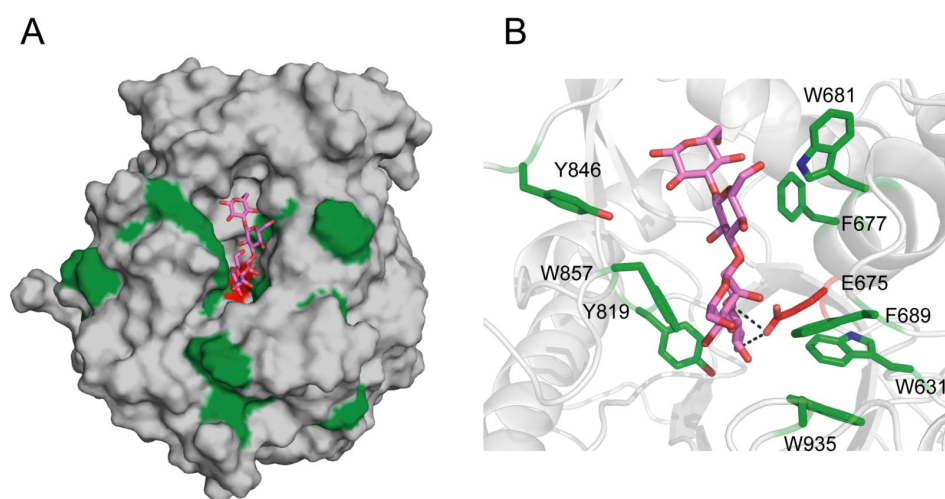
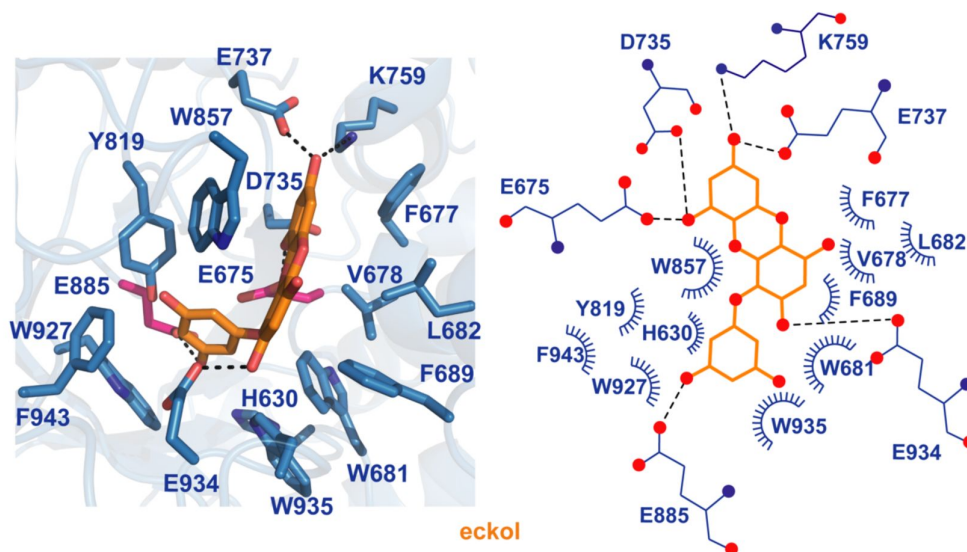


Figure S6.

The model of GH1D2 docking with the substrate laminaritetraose. (A) Overall model of GH1D2-laminaritetraose. GH1D2 is shown as a grey surface representation and laminaritetraose as marine sticks. The aromatic and catalytic residues of GH1D2 are colored green and red, respectively. (B) The closeup review of interaction between laminaritetraose and GH1D2 with key residues showing in sticks.

A



B

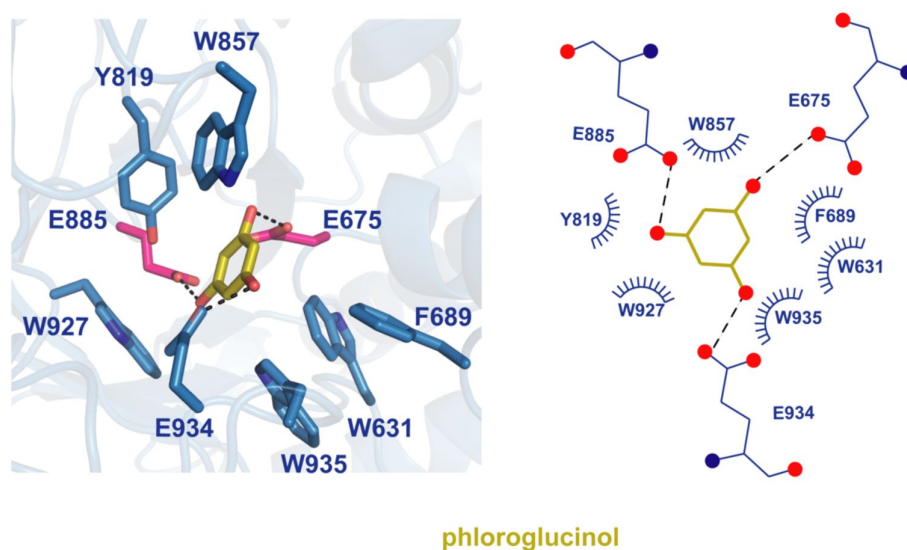


Figure S7.

The models of GH1D2 docking. (A) With eckol, and (B) With phloroglucinol. The left panel shows the 3D structures, and the right panel shows the 2D diagrams. The C, N, and O atoms of residues are colored light blue, dark blue, and red, respectively. The C atoms of eckol and phloroglucinol are shown in orange and yellow, respectively.

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Article and author information

Xiaomei Sun

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Yuxin Ye

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Naofumi Sakurai

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Hang Wang

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Koji Kato

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Jian Yu

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

Keizo Yuasa

Graduate School of Bioscience and Bioindustry, Tokushima University, Tokushima, Japan

Akihiko Tsuji

Graduate School of Bioscience and Bioindustry, Tokushima University, Tokushima, Japan

Min Yao

Faculty of Advanced Life Science, Hokkaido University, Sapporo, Japan

For correspondence: yao@castor.sci.hokudai.ac.jp

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Reviewer #1 (Public Review):

Sun and co-authors have determined the crystal structures of EHEP with/without phlorotannin analog, TNA, and akuBGL. Using the akuBGL apo structure, they also constructed model structures of akuBGL with phlorotannins (inhibitor) and laminarins (substrate) by docking calculation. They clearly showed the effects of TNA on akuBGL activity with/without EHEP and resolubilization of the EHEP-phlorotannin (eckol) precipitate under alkaline conditions (pH >8). Based on this knowledge, they propose the molecular mechanism of the akuBGL-phlorotannin/laminarin-EHEP system at the atomic level. Their proposed mechanism is useful for further understanding of the defensive-offensive association between algae and herbivores.

Reviewer #2 (Public Review):

In this study the authors try to understand the interaction of a 110 kDa β -glucosidase from the mollusk *Aplysia kurodai*, named akuBGL, with its substrate, laminarin, the main storage polysaccharide in brown algae. On the other hand, brown algae produce phlorotannin, a secondary metabolite that inhibits akuBGL. The authors study the interaction of phlorotannin with the protein EHEP, which protects akuBGL from phlorotannin by sequestering it in an insoluble complex.

The strongest aspect of this study is the outstanding crystallographic structures they obtained, including akuBGL (TNA soaked crystal) structure at 2.7 Å resolution, EHEP structure at 1.15 Å resolution, EHEP-TNA complex at 1.9 Å resolution, and phloroglucinol soaked EHEP structure at 1.4 Å resolution. EHEP structure is a new protein fold, constituting the major contribution of the study.

The drawback on EHEP structure is that protein purification, crystallization, phasing and initial model building were published somewhere else by the authors, so this structure represents incremental research.

One concern remains unanswered to me. If the mechanism of action of EHEP is to precipitate together with TNA in a 1:1 insoluble complex, then it does not matter if there are multiple mechanisms involved in the activity assay, the protection of 4 μ M EHEP against 40 μ M TNA simply requires a different stoichiometry.

Reviewer #3 (Public Review):

The manuscript by Sun et al. reveals several crystal structures that help underpin the offensive-defensive relationship between the sea slug *Aplysia kurodai* and algae. These centre on TNA (a algal glycosyl hydrolase inhibitor), EHEP (a slug protein that protects against TNA and like compounds) and BGL (a glycosyl hydrolase that helps digest algae). The

hypotheses generated from the crystal structures herein are supported by biochemical assays.

The crystal structures of apo and TNA-bound EHEP reveals the binding (and thus protection) mechanism. The authors then demonstrate that the precipitated EHEP-TNA complex can be resolubilised at an alkaline pH, potentially highlighting a mechanism for EHEP recycling in the *A. kurodai* midgut. The authors also present the crystal structures of akuBGL, a beta-glucosidase utilised by *Aplysia kurodai* to digest laminarin in algae into glucose. The structure revealed that akuBGL is composed of two GH1 domains, with only one GH1 domain having the necessary residue arrangement for catalytic activity, which was confirmed via hydrolytic activity assays. Docking was used to assess binding of the substrate laminaritetraose and the inhibitors TNA, eckol and phloroglucinol to akuBGL. The docking studies revealed that the inhibitors bound akuBGL at the glycone-binding suggesting a competitive inhibition mechanism. Overall, most of the claims made in this work are supported by the data presented.

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Sun and co-authors have determined the crystal structures of EHEP with/without phlorotannin analog, TNA, and akuBGL. Using the akuBGL apo structure, they also constructed model structures of akuBGL with phlorotannins (inhibitor) and laminarins (substrate) by docking calculation. They clearly showed the effects of TNA on akuBGL activity with/without EHEP and resolubilization of the EHEP-phlorotannin (eckol) precipitate under alkaline conditions (pH >8). Based on this knowledge, they propose the molecular mechanism of the akuBGL- phlorotannin/laminarin-EHEP system at the atomic level. Their proposed mechanism is useful for further understanding of the defensive-offensive association between algae and herbivores. However, there are several concerns, especially about structural information, that authors should address.

Thank you for reviewing our manuscript. We addressed all comments below.

1. TNA binding to EHEP

The electron densities could not show the exact conformations of the five gallic acids of TNA, as the authors mentioned in the manuscript. On the other hand, the authors describe and discuss the detailed interaction between EHEP and TNA based on structural information. The above seems contradictory. In addition, the orientation of TNA, especially the core part, in Fig. 4 and PDB (8IN6) coordinates seem inconsistent. The authors should redraw Fig. 4 and revise the description accordingly to be slightly more qualitative.

We apologize for the mistake with the PDB file. We forgot to re-upload the final coordinate file of 8IN6, which had been modified according to the requirement of the PDB instructions. We have now re-uploaded the correct PDB file. We carefully checked Fig. 4 (Fig.3 in the revised version), which used the final coordinate file of 8IN6.

1. Two domains of akuBGL

The authors concluded that only the GH1D2 domain affects its catalytic activity from a detailed structural comparison and the activity of recombinant GH1D1. That conclusion is probably reasonable. However, the recombinant GH1D2 (or GH1D1+GH1D2) and inactive mutants are essential to reliably substantiate conclusions. The authors failed to overexpress recombinant GH1D2 using the E. coli expression system. Have the authors tried GH1D1+GH1D2 expression and/or other expression systems?

By referencing other BGLs (six samples were expressed by using E. coli, and one was expressed by using Pichia), we only tried the overexpression of akuBGL, GH1D1, GH1D2, and GH1D1+GH1D2 in E. coli expression system using several different vectors. As the reviewer mentioned that inactive mutants are essential to substantiate our conclusion reliably, it will be tried further to use yeast or cell expression systems to confirm our conclusion. We added these limitations as “Future assay of GH1D2 and inactive mutants is the complement to validate the molecular mechanism of akuBGL” in the discussion (Line 343-345)

1. Inhibitor binding of akuBGL

The authors constructed the docking structure of GH1D2 with TNA, phloroglucinol, and eckol because they could not determine complex structures by crystallography. The molecular weight of akuBGL would also allow structure determination by cryo-EM, but have the authors tried it? In addition, the authors describe and discuss the detailed interaction between GH1D2 and TNA/phloroglucinol/eckol based on docking structures. The authors should describe the accuracy of the docking structures in more detail, or in more qualitative terms if difficult.

Yes, it is possible to try cryo-EM for obtaining the structure of akuBGL complexed with the ligand. However, we didn't try because 110 kDa akuBGL consists of two 55 kDa GH1Ds linked by along loop, and we worried that ligand may not be visualized using cryo-EM.

Following the comment, we added the description of the accuracy of the docking structures as “Those docking scores corroborated well with the inhibition activity toward akuBGL, that TNA had a more robust inhibition activity than phloroglucinol, indicating that the docking results are reasonable.” (Line 322-324)

Reviewer #2 (Public Review):

*In this study the authors try to understand the interaction of a 110 kDa β -glucosidase from the mollusk *Aplysia kurodai*, named akuBGL, with its substrate, laminarin, the main storage polysaccharide in brown algae. On the other hand, brown algae produce phlorotannin, a secondary metabolite that inhibits akuBGL. The authors study the interaction of phlorotannin with the protein EHEP, which protects akuBGL from phlorotannin by sequestering it in an insoluble complex.*

The strongest aspect of this study is the outstanding crystallographic structures they obtained, including akuBGL (TNA soaked crystal) structure at 2.7 Å resolution, EHEP structure at 1.15 Å resolution, EHEP-TNA complex at 1.9 Å resolution, and phloroglucinol soaked EHEP structure at 1.4 Å resolution. EHEP structure is a new protein fold, constituting the major contribution of the study.

We thank you for reviewing our manuscript.

The drawback on EHEP structure is that protein purification, crystallization, phasing and initial model building were published somewhere else by the authors, so this structure is incremental research and not new.

We have published the results of protein purification, crystallization, phasing, and initial model building for determining structure but have yet to give the structure since further structural refinement is indispensable. Such published data in [Acta F] is a service for obtaining the structure.

We believe that the structure of the EHEP holds great importance, and it is the first time to publish.

Most of the conclusions are derived from the analysis of the crystallographic structures. Some of them are supported by other experimental data, but remain incomplete. The impossibility to obtain recombinant samples, implying that no mutants can be tested, makes it difficult to confirm some of the claims, especially about the substrate binding and the function of the two GH1Ds from akuBGL.

As mentioned by the reviewer, mutant analysis would be the best way to substantiate our conclusions. However, it is challenging to obtain recombinant samples, although we tried to overexpress them (akuBGL, GH1D1, GH1D2, and GH1D1+GH1D2). So, we did the structural comparison, and docking simulation to propose the molecular mechanism. We added these limitations as "Further assay of GH1D2 and inactive mutants is the complement to validate the molecular mechanism of akuBGL" in the discussion part (Line 343-345).

The authors hypothesize from their structure that the interaction of EHEP with phlorotannins might be pH dependent. Then they succeed to confirm their hypothesis, showing they can recover EHEP from precipitates at alkaline pH, and that the recovered EHEP can be reutilized.

A weakness in the model is raised by the fact that the stoichiometry of the complex EHEP:TNA is proposed to be 1:1, but in Figure 1 they show that 4 μ M of EHEP protects akuBGL from 40 μ M TNA, meaning EHEP sequesters more TNA than expected, this should be addressed in the manuscript.

The assay experiment in figure1 does not directly provide the stoichiometric ratio of EHEP:TNA because the activity assay system consists of substrate of akuBGL, akuBGL, TNA, and EHEP, which involves multiple equilibration processes: $\text{akuBGL} \rightleftharpoons \text{substrate}$, $\text{akuBGL} \rightleftharpoons \text{TNA}$, and $\text{EHEP} \rightleftharpoons \text{TNA}$. To avoid misunderstanding, we added the descriptions of "As this activity assay system involves multiple equilibration processes: $\text{akuBGL} \rightleftharpoons \text{substrate}$, $\text{akuBGL} \rightleftharpoons \text{TNA}$, and $\text{EHEP} \rightleftharpoons \text{TNA}$." (Line 120-121).

The authors study the interaction of akuBGL with different ligands using docking. This technique is good for understanding the possible interaction between the two molecules but should not be used as evidence of binding affinity. This implies that the claims about the different binding affinities between laminarin and the inhibitors should be taken out of the preprint.

Following the suggestion, we deleted the descriptions about the difference in binding affinity with docking scores at the last paragraph of [Inhibitor binding of akuBGL].

In the discussion section there is a mistake in the text that contradicts the results. It is written "EHEP-TNA could not dissolve in the buffer of pH > 8.0" but the result obtained is the opposite, the precipitate dissolved at alkaline pH.

We apologize for this mistake and corrected it to "EHEP-TNA could dissolve in the buffer of pH > 8.0." (Line 394).

Solving a new protein fold, as the authors report for EHEP, is relevant to the community because it contributes to the understanding of protein folding. The study is also relevant due to the potential biotechnological application of the system in biofuel production. The understanding on how an enzyme as akuBGL can discriminate between substrates is important for the manipulation of such enzyme in terms of improving its activity or changing its specificity. The authors also provide with preliminary data that can be used by others to produce the proteins described or to design a strategy to recover EHEP from precipitates with phlorotannin at industrial scales.

In general methods are not carefully described, the section should be extended to improve the manuscript.

Following the comment, we added the method descriptions

1. Recombinant GH1D1 domain expression and purification in [EHEP and akuBGL preparation].
2. Sections of [recomGH1D1 activity assay], and [N-terminal sequencing of akuBGL]
3. More details of resolubilization of EHEP and activity in [Resolubilization of the EHEP-eckol precipitate].

Reviewer #3 (Public Review):

*The manuscript by Sun et al. reveals several crystal structures that help underpin the offensivedefensive relationship between the sea slug *Aplysia kurodai* and algae. These centre on TNA (a algal glycosyl hydrolase inhibitor), EHEP (a slug protein that protects against TNA and like compounds) and BGL (a glycosyl hydrolase that helps digest algae). The hypotheses generated from the crystal structures herein are supported by biochemical assays.*

*The crystal structures of apo and TNA-bound EHEP reveals the binding (and thus protection) mechanism. The authors then demonstrate that the precipitated EHEP-TNA complex can be resolubilised at an alkaline pH, potentially highlighting a mechanism for EHEP recycling in the *A. kurodai* midgut. The authors also present the crystal structures of akuBGL, a beta-glucosidase utilised by *Aplysia kurodai* to digest laminarin in algae into glucose. The structure revealed that akuBGL is composed of two GH1 domains, with only one GH1 domain having the necessary residue arrangement for catalytic activity, which was confirmed via hydrolytic activity assays. Docking was used to assess binding of the substrate laminaritetraose and the inhibitors TNA, eckol and phloroglucinol to akuBGL. The docking studies revealed that the inhibitors bound akuBGL at the glycone-binding suggesting a competitive inhibition mechanism. Overall, most of the claims made in this work are supported by the data presented.*

We thank you very much for reviewing our manuscript.

Reviewer #1 (Recommendations For The Authors):

- Fig. 3 should be moved to the Supplements because acetylation modification at the N-terminus is not essential for the function of EHEP.

Following the recommendation, we moved Fig.3 to Supplements (Fig. S2).

- EHEP2 is processed at 1.4 Å resolution, however, the statistics at highest resolution shell indicate you can process at higher resolution. Why 1.4 Å resolution?

We tried to process this dataset at the higher resolution at 1.35 Å, and the completeness and I/sigma of the highest resolution shell reduced to 88.9% and 2.16, respectively. The parameter of I/sigma is OK, but the completeness reduced seriously. So, we set a cutoff of 1.4 Å.

- Fig. S1A should be revised to include the gallic acid numbers (1, 2, 3, 4, 6) and the 3.0 σ map. >

As presented in Fig. S1A, the omitted map (fo–fc map) of the ligand TNA, countered at 2.0 σ , showed that gallic acid 2 has poor density, and gallic acid 4 has weak density. Moreover, the TNA is relatively big to EHEP (7.5 %), and the omitted map countered 3.0 σ could not clearly show gallic acids. So, we keep the map at 2.0 σ in Fig. S3A.

- The authors should provide more information on "co-cage-1 nucleant".

Our lab is currently publishing a paper that provides detailed information on the co-cage-1 nucleant, including components, synthesis, nucleation mechanism, and application. Once the paper is published, we will cite it in this manuscript.

Reviewer #2 (Recommendations For The Authors):

- Is the word "offence" the appropriate word for referring to the activity of EHEP? Is this word used in the literature for this system? I find it confusing but might be because I am not in the specific topic.

In the field of prey–predator, the defense–offensive is commonly used. According to Charles D. Amsler's book "Algal Chemical ecology", Herbivore offensive is the traits that allow herbivores to increase feeding rates on algae. Therefore, in our opinion, the offensive is appropriate.

Taking into consideration that I am not an English language expert I find the writing of the manuscript could be improved in general. Here are some lines as examples of where the grammar could be better:

Line 193: "decrement of the loop part"

Following the comment, we corrected it to "decrease of the loop part" (Line 197).

Line 199: there is a typographical error.

We apologize for our mistake and corrected it to "EHEP" (Line 202).

Line 205-206: "only hydrophobically interacted with"

Following the comment, we modified it to "only interacted hydrophobically with EHEP" (Line 209)

Line 224: "phlorotannin-precipitate activity"

Following the comment, we modified it to "phlorotannin-precipitate activity" (Line 227).

| *Line 232: "without the N-terminal 25 residues"*

Following the comment, we modified it to "lacked the N-terminal 25 residues" (Line 236).

| *Line 353: "bound" should be "bind"*

We apologize for our mistake and modified it (Line 356).

| *Line 359: "predator mammals"*

We apologize for our mistake and modified it to "predatory mammals" (Line 363).

| *Line 363: "at an alkaline pH of insect midgut"*

Following the comment, we modified it to "at the alkaline pH of the insect midgut" (Line 367).

| *Line 370: "nonstructural proteins" means "unstructured proteins"?*

Yes, unfolding proteins, we modified to "unfolding proteins with randomly coils" (Line 374).

| *Line 374: "similar strategy with mammals"*

Following the comment, we modified it to "similar strategy to mammals" (Line 379).

| *Line 403: "to forming"*

We apologize for our mistake and modified it to "to form" (Line 404).

| *Line 404: "considered no binding"*

We apologize for our mistake and modified it to "considered not binding" (Line 405).

| *Line 406: "activity pocket" means the active site?*

Yes, we modified it to "active site" (Line 407).

| *Line 424: "step purification"*

Following the comment, we corrected it to "one step for purification" (Line 425).

| *Line 431*

Following the comment, we corrected it to "To verify whether the chemical modifications which was indicated by previous study affects" (Line 432-433).

| *Line 812: there is typographical error*

We apologize for our mistakes, and corrected it to Tris-HCl" for all "Tris-HCl (Line 878~).

| *Line 223: eckol is not mentioned in the text and appears for the first time in the figure caption.*

Following the comment, we added "eckol" in the first section of the [Result] (Line 117).

The paragraph between lines 271 and 280 is disconnected from the previous one and it is not about results, it should be at the discussion section.

Following the comment, we moved them to the discussion part (Line 335-343).

Line 324: "the three inhibitors inhibited": this claim should be corrected to "the three inhibitors interacted", since the word inhibited would imply the authors measured activity experimentally.

We modified it as the comment. (Line 325).

Line 392: "could not dissolve" is contradicting the result.

We apologize for our mistake and corrected it to "could dissolve" (Line 394).

They describe acetylation but they try overexpressing in E. coli, could it be that they needed to express the construct in a system where they would get the acetylation? At least this should be discussed in the text.

Because our sample of EHEP with acetylation was purified from the natural source of the digestive fluid of *A. kurodai*, we only need to express EHEP without acetylation. Following the comment, we modified the descriptions to clarify it in the section (Lines 170-173 and 177-179).

"Consistent with the molecular weight results obtained using MALDI-TOF MS, the apo structure2 (1.4 Å resolution) clearly showed that the cleaved N-terminus of Ala21 underwent acetylation, demonstrating that EHEP is acetylated in *A. kurodai* digestive fluid."

"To explore whether acetylation affects the protective effects of EHEP on akuBGL, we used the *E. coli* expression system to obtain the unmodified recomEHEP (A21-K229)."

From the text it is not clear in which biological context the brown algae meet the attack by the hydrolase, the information is spread all over the manuscript, it should be clearly described at the introduction.

When the brown algae are consumed as food by sea hare *A. kurodai*, they meet the attack by the hydrolase akuBGL. Following the comment, we clear the descriptions in the introduction part as below (Line 42-45).

"In brown algae *Eisenia bicyclis*, laminarin is a major storage carbohydrate, constituting 20%–30% of algae dry weight. The sea hare *Aplysia kurodai*, a marine gastropod, preferentially feeds on the *E. bicyclis* with its 110 and 210 kDa β -glucosidases (akuBGLs), hydrolyzing the laminarin and releasing large amounts of glucose."

Affinity ranking based on docking is not reliable, the differences in free energy are in the same order of magnitude. I would recommend erasing this claim since it is not fundamental to the study. Another option would be to determine affinities experimentally.

We agree with the comment and removed the text about affinity ranking with docking scores.

Figure 1: relative activity is not defined. HPLC data should be shown as supplementary material.

Following the comment, we added the definition of relative activity and the HPLC data as Fig. S1 in the revised version.

Figure 4: Sephacryl resin is mentioned here but not described in the methods.

Following the comment, we added the description in the methods (Line 515).

Protein N-terminal sequencing analysis should be described in the methods.

Following the comment, we added the sequencing analysis in the methods (Line 476-483).

Figure S1 C: it should be specified how the surface electrostatic potential at different pH was calculated.

Following the comment, we added the descriptions of how the surface electrostatic potential at different pH was calculated in the figure legend of Fig. S2 of the revised version (Line 876-877).

Since the authors are capable of producing good amounts of akuBGL and have already conducted glycosidase activity assays using ONPG, it would not be difficult for them to run some kinetics experiments for the enzyme in the presence of the different inhibitors to confirm their hypothesis derived from the docking calculations.

As mentioned by the reviewer, kinetics experiments are the best way to confirm our hypothesis derived from docking calculations. However, the yield of akuBGL purification from the digestive fluid of sea hare *A.kurodai* is quite difficult. We could not obtain a sufficient sample of akuBGL to conduct the kinetic experiments. So, we stopped at docking simulation in this study. We added such limitations of "Future kinetic experiments are required to validate quantitatively the competitive inhibition of phlorotannin against akuBGL" (Line 359-360).

Some citations are missing in the discussion section, for example in lines 362, 364 and 396.

Following the comment, we added the citations.

Reviewer #3 (Recommendations For The Authors):

Please see comments/suggestions below for revisions.

Line 176-178 - Text explains that recomEHEP precipitated after incubation with TNA to a comparable level to natural EHEP. However, figure 3B shows no comparison between recombinant and natural EHEP.

As the reviewer suggested, we repeated the binding assay of recomEHEP to confirm the precipitation with TNA and added a precipitation result of natural EHEP (Fig. S2B right) for comparing.

Line 223 - The work presented in Figure S1E goes partway towards demonstrating the activity of resolubilised EHEP. This claim would be strengthened if resolubilised EHEP was used in the akuBGL Galactoside hydrolytic activity assay and is then seen to rescue akuBGL activity in the presence of TNA.

Yes, our claim would be strengthened by adding resolubilized EHEP to akuBGL assay in the presence of TNA. Since we have obtained and presented the relationship between the precipitating of EHEP with TNA and the rescuing akuBGL activity from TNA, we only used the precipitation to demonstrate the activity of resolubilized EHEP.

Line 380-384 - Here it is discussed how TNA simultaneously binds to three EHEP molecules thus crosslinking them. It is then proposed that this could be the mechanism of precipitation. However, it is noted that TNA is soaked into crystals, therefore it is likely that this lattice exists whether TNA is present or not (this absolutely needs to be mentioned in the text). It would be possible to test this mechanism through mutagenesis. If the sites where TNA packs in between chains of EHEP were mutated to prevent crosslinking, it could then be determined whether crosslink-null EHEP can still precipitate TNA.

As the review mentioned, we do not have enough experiments to propose that the TNA-crosslink may cause the EHEP-TNA precipitation. So, we deleted the discussion of the TNA crosslink and the corresponding figure.

All docked models need to be deposited (perhaps modelarchive.org) and this resource referred to in the text.

The structures in modelarchive.org site are either homology models or de novo. We think the docked model is out of this site. So, we did not deposit them.

The x-ray data table contains data previously published in the referenced Acta crystallographica publication. What is eLife policy on this "double use" of data?

We apologize for our mistake, and deleted the SAD data in Table 1.

Minor points

Line 26 - use "apo akuBGL" so as not to infer a tannic-acid bound form of this also >

Following the comment, we modified it to "apo akuBGL" (Line 26).

Line 48 - The sentence currently reads as A. kurodai is being digested.

Following the comment, we modified it to "by A. kurodai" (Line 48).

Line 49-50 & Line 65-66 - Both these lines make the same point about the impact of phlorotannin inhibition on the use of brown algae as feedstocks for biofuel, please remove one.

Following the comment, we deleted the line 49-50.

Line 115 - This needs attention as its an unusual opening sentence

Following the comment, we modified it to "Phlorotannin, a type of tannin, is a chemical defense metabolite of brown algae." (Line 114).

Line 130 - Should the EHEP concentration be 3.96 μ M not 3.36?

We apologize for our mistake 3.36 is correct, and we corrected the X-axis label in Fig.1B.

Line 133 - consider using "non-recombinant" rather than "natural"

To distinguish between non-recombinant and recombinant samples, we used “EHEP” and “akuBGL” as purified from the native source and recomEHEP and recomakuBGL as the samples overexpressed from *E. coli* in this manuscript. So, we added the definition in [Introduction] (Line 100-101).

Line 134 - "The residues A21-V227 of A21-K229..." This sentence could be written more clearly.

Following the comment, we re-wrote it to “The residues A21–V227 in purified EHEP (1–20 aa were cleaved during maturation) were built” (Line 135-136).

Line 136 - switch "appropriately visualized" for "tracable"?

Following the comment, we modified it to “built” (Line 136).

Line 158 - use "70% of backbone in a loop conformation" >

We modified as the comment (Line 159-160).

Line 184 - reword "map showed an electron density blob". (Map showed positive electron density)

Following the comment, we modified it to “map showed the electron density” (Line 188).

Line 193-194 - Is EHEP really more stable when bound to TNA? It is not shown experimentally? It is difficult to see which loop changes. Is the difference a result of crystal packing? Please switch "decrement" for another term

The regions with conformation change between EHEP and EHEP–TNA are close to TNA but not at the intermolecular interface. As the reviewer mentioned, we could not clarify the EHEP stability depended on TNA-binding, and deleted the descriptions in the second paragraph of [TNA binding to EHEP].

Following the comment, we redraw Fig. S1B (Fig. S3B in the revised version) to show the conformation changes clearly. We also modified “decrement” to “decrease” (Line 197).

Fig S1B - Can an extra figure be added to show the secondary differences more clearly? >

We redraw this figure (Fig. S3B) using closeup view to show the differences.

Line 212-213 - There is a slight discrepancy between the text and Figure 4B. Gallic acid 4 interacts with P201 and gallic acid 6 interacts with P77.

We apologize for our mistake in the text. and corrected it to “gallic acid4 and 6 showed alkyl– π interaction with P201 and P77, respectively” (Line 216).

Figure 4D - Change x axis from tube number to elution volume. Both chromatograms could also be superimposed for interpretability.

Since we used raw data from the experiment, we kept the x-axis in tube number with additional “2.7 ml/tube” information (Fig.3D).

Line 229 - Please change "there was no blob of TNA in the electron density" to there was no electron density for TNA or something similar.

Following comment, we modified it to “there was no electron density of TNA or something similar in the 2Fo–Fc and Fo–Fc map” (Line 232).

Line 231 - asymmetric unit is a more standard term (also in Fig S2 legend)

We modified as the comment (Line 235 and 885).

Line 234-235 - Reword "the residues L26-P978 of L26-N994" to make it more concise. >

Following the comment, we deleted “of L26-N994” (Line 239).

Lines 296-299 could be written more carefully - pi stacking with what? >

We apologize for our mistake and corrected it to CH– π (Line 293).

Line 349 - which putatively enables it to.....

We modified it as the comment (Line 353 in the revised manuscript).

Line 370 - "nonstructural" is the wrong term because they remain structured - use something akin to non-classical secondary structure

Following the comment, we modified it to “are unfolding proteins with randomly coils in solution ” (Line 374)

Throughout - use phenix autobuild, not autobuil

We apologize for our mistakes and corrected them throughout the manuscript.

Figure 1 - the graphs would be more interpretable with all data points shown overlaid

The two graphs in Figure 1 showed two experiments with different reaction conditions. Figure 1A presents various TNA concentrations, while Figure 1B maintains a constant concentration of 40 μ M for TNA with varying EHEP concentrations. So, overlaying the graphs is not feasible. Therefore, we would like to keep them separated and added the reaction condition in figure legend.

Figure 4 - in part D add an extra statement outlining what the S-100 analysis demonstrated

S-100 analysis is using a gel filtration column with Sephacryl S-100 media. We added an extra statement in the method and the legend (Fig. 3, Lines 515 and 879).

Figure 5 (and elsewhere) - the structures referred to need a PDB code and reference given in legend

Following the comment, we checked the manuscript carefully and added PDB code to the referred structures.

Fig S1 - please add an additional panel showing part D but in proper structure form, not schematic shapes

Since we do not have enough experiments to validate the TNA-crosslink, we deleted the discussion of the TNA crosslink and Fig. S1D.

Figure sig 4 - Text contains in depth information of side chain hydrogen bonding and π - π interactions between akuBGL and laminarittrose. However, the figure only shows a surface model. Consider adding a figure showing these interactions.

Following the suggestion, we added a closeup view to show these detailed interactions (Fig. S6B).